# NEW LOWER BOUNDS FOR KISSING NUMBERS IN DIMENSIONS 18 THROUGH 96

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ABSTRACT. The kissing number $K(n)$ is the largest number of unit vectors in $\mathbb{R}^n$ with pairwise inner products at most $\frac{1}{2}$. Specialized search systems recently improved the eleven-dimensional records established by Ganzhinov and AlphaEvolve on EinsteinArena. We therefore asked whether commonly available language models and agentic coding tools, including Claude Code and OpenAI Codex, could contribute mathematical advances. Surprisingly, a single Claude Opus 5 session, after some motivational prompting, produced a new dimension-27 construction. Scaling the workflow with Claude Code and Codex led to 65 certified lower-bound constructions in dimensions 18–96, with 59 still state of the art in our review. Several exceed records reported by specialized search systems, including the Qiushi Engine. On the broader leaderboard, Li reports GPT-6 Astra-assisted records in dimensions 19 and 21 and later lifted-block records in dimensions 28–31. Where our earlier entries have since been superseded, Takhanov–Yun's AI-generated constructions improve dimensions 25–27, 29 and 31, while Lin's AI-assisted code and equator searches supply later records in dimensions 32–35 and 37–39.

Most constructions come from the extremal even unimodular lattices $\Lambda_{24}$, $P_{48}$ and $\Gamma_{72}$, through lattice cross-sections, lifted layers and code constructions; the remaining entries include a bent-coset construction and evaluations of the Edel–Rains–Sloane chain. Every bound is certified in exact arithmetic. The continuing gains suggest that these lower bounds are far from saturated, especially in higher dimensions, where we expect substantially more progress with additional compute and more extensive prompting of Codex. Their open-ended geometry makes kissing-number lower bounds a useful training and evaluation benchmark for LLMs and agents: every candidate must pass an exact geometric test, while higher dimensions provide fresh instances beyond any finite list of memorized configurations.

## 1. INTRODUCTION

The *kissing number* $K(n)$ is the largest number of non-overlapping unit balls that can touch a central unit ball in $\mathbb{R}^n$. Equivalently, it is the largest

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size of a set of unit vectors in $\mathbb{R}^n$ whose pairwise inner products are at most $\frac{1}{2}$. Such a set is a *kissing configuration*, or a spherical code of minimum angle $60°$.

Exact values are known only for $n = 1, 2, 3, 4, 8, 24$. The cases $n = 3$ [45], $n = 4$ [37], $n = 8$ and $n = 24$ [42, 29] are hard theorems. In every other dimension the known bounds leave an interval whose two ends move independently. This paper concerns the lower end.

1.1. **Motivation and workflow.** Specialized search systems have recently improved the eleven-dimensional records established by Ganzhinov and AlphaEvolve on the EinsteinArena platform [20, 41, 2]. This prompted us to test commonly available language models and agentic coding tools on the same problem. Surprisingly, a single Claude Opus 5 session, after some motivational prompting, produced the dimension-27 construction described in Appendix F.3. We then scaled the work with Claude Code and OpenAI Codex, using them for exploration, implementation and verification while retaining human selection of the mathematical questions and exact certification of every stated bound. The resulting constructions improve records in several dimensions where specialized systems had recently reported advances, including values associated with the Qiushi Engine [51]. Independent recent work by Li, using GPT-6 Astra among other tools, reports further improvements in dimensions 19, 21 and 28–31 [30, 31]. The lower-bound problem is productive without appearing saturated: an agent can find a better construction, while higher dimensions and alternative geometries keep creating new targets. We expect substantially more progress in those dimensions as compute budgets grow and Codex is prompted more extensively. This makes kissing-number lower bounds a useful training and evaluation setting for LLMs and agents. Each proposed configuration must pass an exact geometric test, and the unbounded range of dimensions makes it hard to reduce the task to a finite list of memorized examples.

1.2. **The published tables.** The state of the art is Cohn's *Table of kissing number bounds* [6, 7], which supersedes the Nebe–Sloane table [40]. Their ranges fix the meaning of "previously known" below. Cohn's table covers dimensions 1–48 and 72; the Nebe–Sloane table covers 1–40, 42, 44, 48, 64, 72, 80 and 128. Above dimension 48 the two together have entries at $n = 64$, 72, 80 and 128 and at no other dimension.

In dimensions 49–63 the best published value therefore comes from the orthogonal direct sum of the minimal shell of $P_{48}$ with a kissing configuration of $\mathbb{R}^{n-48}$ and equals $52\,416\,000 + K(n-48)$, where $K(n-48)$ is Cohn's entry in dimension $n-48$, the argument running from 1 to 15. In dimensions 65–71 it is 331 737 984, inherited by monotonicity of $K$ from the dimension-64 entry of Edel–Rains–Sloane [19]. In dimensions 73–96 the same construction over $\Gamma_{72}$ gives $6\,218\,175\,600 + K(n-72)$, which at $n = 96$ is $6\,218\,175\,600 + 196\,560$, the second summand being the Leech lattice; here $K(n-72)$ is again Cohn's

entry in the historical comparison snapshot, its argument running from 1 to 24. Many current entries lie in these ranges.

The only dimension inside those three ranges that is tabulated is $n = 80$, where the Nebe–Sloane record $\Gamma_{72} \perp E_8$, whose kissing number $6\,218\,175\,600+240$ is the direct sum above. The rule and the table agree there, and that is the only place they meet.

A table entry is not the only thing a bound above dimension 48 has to beat. The construction of Edel–Rains–Sloane is generic in $n$, and evaluated in the dimension itself it can exceed the value inherited from a smaller one: in dimensions 65–71 it does, by a little, giving $331\,771\,800$ at $n = 68$ (Remark B.13). The "previously" column of Table 1 is the tabular floor defined here; §6.1 measures every bound of this paper against the constructive one, which is the stronger test; within the cross-section programme, the dimension-68 calculation is the tightest comparison.

1.3. **Cross-sections and layers.** Both constructions start from the minimal shell of one of the three extremal even unimodular lattices

$$\begin{array}{lll} \Lambda_{24} & n = 24,\ \mu = 4, & N = 196\,560, \\ P_{48} & n = 48,\ \mu = 6, & N = 52\,416\,000, \\ \Gamma_{72} & n = 72,\ \mu = 8, & N = 6\,218\,175\,600, \end{array}$$

where $\mu$ is the minimum and $N$ the number of minimal vectors. The minimal shell of each, rescaled to the unit sphere, is a kissing configuration; $\Lambda_{24}$ attains $K(24) = 196\,560$.

The first construction takes cross-sections. The minimal vectors orthogonal to $k$ fixed minimal vectors span a subspace of dimension at most $n - k$ and have pairwise inner products at most $\frac{1}{2}$; they therefore form a kissing configuration in $\mathbb{R}^{n-k}$. The problem is to count them, for which Venkov's theorem provides the constraints (§B).

The second construction places the shell on an equator of $\mathbb{R}^n \oplus \mathbb{R}^k$ and adjoins copies of part of it, lifted out of the equatorial subspace above a small set of directions in $\mathbb{R}^k$ (§C). Recent constructions in dimensions 25 through 31 develop this lattice-layer approach.

A cross-section loses points and gives a bound below dimension $n$; the layered construction gains points and gives one above it. The same quantity limits both: how many minimal lines of $L$ can be pairwise at small inner product.

1.4. **Results.** Table 1 gives one row for each of the 59 bounds that remain state of the art in the reviewed sources. It reports the recorded comparison, the current value, and the construction family; the appendix gives the construction data and proofs, while Section 6.3 describes the exact certification. Six superseded claims are retained for reproducibility; their descriptions and the September manuscript table appear in Appendix F.

Shared midpoint replacements give 200 918 and 204 636 in dimensions 26 and 27, regaining both records after Takhanov–Yun. The current additions

are concise by design. The Coxeter–Todd construction gives 501 633 and 505 359 in dimensions 36 and 37; the $P_{48n}$ construction supplies dimensions 41–46; the code-layer constructions give 71 310 488, 100 371 468, 182 407 428, 332 170 112, 332 238 338, 338 292 100, 351 400 326 and 371 521 432 in dimensions 61–68, and 7 091 452 684 and 13 289 406 656 in dimensions 95 and 96; the nearest-coset construction supplies dimensions 73–86, and the compact cap certificates supply dimensions 87–94.

1.5. **Attribution.** *Dimensions* 46 *and* 47. The value 23 766 960 is due to Boyvalenkov–Cherkashin [3]: it is equation (3) of the preprint `arXiv:2312.05121`, whose surrounding paragraph observes that those $A_0$ vectors form a kissing configuration in $\mathbb{R}^{47}$. The value 12 309 600 is Table 3 of Sun–Wang [46] (`arXiv:2607.20359v3`), obtained by enumerating the 52 416 000 minimal vectors of $P_{48}$; the degree-3 Siegel theta tables of Ozeki [44] contain the ingredients a decade earlier. We derive both in §B.7 as a check on the cross-section programme, but claim neither. Cohn's table [6] records 5 318 060 in dimension 46 and 9 741 412 in dimension 47.

*Attribution and status.* The table records the current numerical status, while the appendix preserves the earlier constructions and their dates. The one-point distribution of the minimal shell of $\Gamma_{72}$ follows from Venkov's theorem, while dimensions 62, 63 and 96 evaluate the Edel–Rains–Sloane construction against current code tables. Its seven worked examples are dimensions 32, 36, 40, 44, 64, 80 and 128; the range "32 to 128" in that paper's abstract is the span of those examples, and it tabulates no value by dimension.

1.6. **Organisation.** §2 places this work in the literature. §3 fixes notation and states the two classical inputs. Appendix B develops the cross-section method and proves the $\Gamma_{72}$ bounds; Appendix C gives the common layered construction and the current $P_{48}$ and $\Gamma_{72}$ families. Appendix D treats the code-theoretic entries and Appendix E treats dimension 18. §5 collects the current state-of-the-art table, while Appendix F retains the detailed superseded constructions and the frozen September table. §6 records comparisons, remaining questions and verification.

A public software package [26] verifies every bound below, to the standards set out in §6.3; no floating-point computation decides any of them.

## 2. Related work

2.1. **Upper bounds.** The results below concern lower bounds. The upper end is dominated by linear and semidefinite programming. The method of Delsarte [15], in the spherical form of Delsarte, Goethals and Seidel [16], yields the classical bounds of Odlyzko and Sloane [42] and of Levenshtein [29], which determine $K(8)$ and $K(24)$. Semidefinite relaxations strengthen it. Bachoc and Vallentin [1] introduced the three-point bound; Mittelmann and Vallentin [36] computed it to high accuracy; Machado and de Oliveira

Filho [35] reduced it by polynomial symmetry. The clustered low-rank solvers of Leijenhorst and de Laat [28] produce most of the upper-bound column of [6] from dimension 10 upwards and establish the optimality of the $D_4$ root system [14]. The gap remains wide in the dimensions treated here: in dimension 47 the best known construction gives 23 766 960 (§B.7) against an upper bound of 621 658 419.

2.2. **Sphere packing.** The kissing number problem is the local form of the sphere packing problem. The two have shared techniques since Cohn and Elkies [8] introduced the linear programming bound for packing density. Viazovska [50] settled the packing problem in dimension 8, and Cohn, Kumar, Miller, Radchenko and Viazovska [10] in dimension 24; the same authors later proved universal optimality of $E_8$ and $\Lambda_{24}$ [11]. Each of these proofs exhibits a modular form making the linear programme tight. Dimensions 8 and 24 are exactly the dimensions in which the linear programming bound already determines the kissing number. No such form is available in the dimensions considered here.

2.3. **Constructions from codes.** The Edel–Rains–Sloane construction [19] assembles a $60^\circ$ code from a chain of binary constant-weight codes. Write $A(n, d)$ for the largest size of a binary code of length $n$ and minimum distance $d$, and $A(n, d, w)$ for the corresponding constant-weight quantity. Improved code tables therefore give improved kissing configurations without changing the construction. Echols [18] supplies 124 new constant-weight codes. Lin [32, 33] gives explicit certificates for 352 320 points in dimension 32, and 370 240, 390 980, 415 308, 499 816, 605 322 and 768 916 in dimensions 33, 34, 35, 37, 38 and 39. Olivier [43] subsequently raises the dimension-37 value to 500 008. Sun and Wang [46], and Dorofeev, Sun and Wang [17], develop cross-section and antipode constructions that provide the comparison values in dimensions 40–46. Yang et al. [51] report a broad autonomous-search programme covering 25, 27, 32–39, 43, 45 and 49–55.

2.4. **Algebraic constructions.** Ganzhinov [20] constructs highly symmetric line systems from twisted spherical functions of finite groups and obtains 510, 592 and 1932 points in dimensions 10, 11 and 14. The dimension-14 configuration provides the directions used in Appendix F.6. Cohn and Li [12] improve dimensions 17 through 21 using odd-sign configurations and punctured Golay codes. Ho [23] raises the dimension-19 value to 11 948, and Li [30] subsequently raises the public lower bounds to 12 460 in dimension 19 and 31 560 in dimension 21. The former uses a hyperplane section and reshaped projected sign vectors; the latter uses Golay sign vectors with unequal coordinate magnitudes. Zinoviev and Ericson [53] give an earlier small-dimension construction based on designs. For dimensions 25–31, Kallal, Kan and Wang [24] and Ma et al. [34] provide earlier Leech-layer constructions. Takhanov and Yun [48] later improve 25–27, 29 and 31, while Li [31] gives the larger lifted-block records in 28–31.

2.5. **Numerical and machine-assisted search.** A separate line of work obtains configurations by numerical optimisation, and several of the values improved on below come from it.

Ma et al. [34] supply important predecessor configurations in dimensions 25–31. Certificate-driven improvements by Takhanov and Yun and by Li are described above. Takhanov, Assylbekov and Yun [47] obtain $K(12) \geq 841$ by energy minimisation initialised from an analysis of the known 840-point arrangements.

Early evolutionary search over programs driven by large language models improved dimension 11. AlphaEvolve [41] raised $K(11)$ from 592 to 593; agents on the EinsteinArena platform [2] later raised it to 604. Georgiev, Gómez-Serrano, Tao and Wagner [21] applied AlphaEvolve to 67 problems and examined the kissing number in dimensions 1 through 11. They recover the known lower bound in each dimension and report 593 in dimension 11, adding no further improvement; dimension 12 was not attempted. GigaEvo [25] attempted dimension 12 with the aim of improving on the lower bound 840 and reached 840.

The same means have been applied to a related problem. In [27], by Kravatskiy with Khrulkov and Oseledets, the object of study is the spherical code, or Tammes, problem: for fixed $N$ and $d$, arrange $N$ points on $S^{d-1}$ so as to minimise the maximum pairwise inner product. That work samples 90 configurations at random from the catalogue of best known spherical codes maintained by Cohn [5], in dimensions 8 through 16. It lowers the maximum inner product for the majority of them, by up to 2.4% in relative terms, and reports no kissing number result; the question was not asked there.

Yang et al. [51] use the Qiushi Engine research system for coordinated shell motion, support-code changes, signed-vector replacements, second-shell equators, lattice lifts and moment certificates. Their 36-, 43- and 45-dimensional values supply comparisons for our current constructions.

The recent sources also disclose AI assistance. Li [30] lists GPT-6 Astra Pro in ChatGPT, Claude Opus 5.5 and Claude Fable 5.1; Takhanov and Yun [48] state that all their new constructions were generated with Claude Code using Claude Opus 5.5 and GPT-6 Astra. They describe using these systems to rotate additional coordinates and fit more points in dimensions 29 and 31, and to improve the existing records in dimensions 25–27. Lin's AI-assisted code and equator searches supply the later records in dimensions 32–35 and 37–39; the public archives give the exact search and certificate details but do not identify a model. Olivier [43] reports using Claude Opus 5.5 in Claude Code for search, computations and review. These are the authors' reported workflows; results are attributed to their human authors.

2.6. **Scope of the present work.** This paper reports explicit lower bounds from lattice sections, lifted layers, code constructions and nearest cosets. The main table uses the current results and the dated comparison sources above; the appendix retains earlier constructions with their historical status.

The new configurations range from 8 358 points in dimension 18 to more than thirteen billion in dimension 96. Their algebraic and combinatorial descriptions permit exact verification without storing every pair of points.

## 3. Preliminaries

Throughout, $L$ is an even unimodular lattice of dimension $n$ with minimum $\mu$, and $\Lambda \subset L$ is its set of minimal vectors, or *minimal shell*, with $N = |\Lambda|$. For $v, w \in \mathbb{R}^n$ we write $\langle v, w\rangle$ for the standard inner product and $\hat{v} = v/|v|$.

We write $\Lambda_k$ for the laminated lattice of dimension $k$, following [13, Ch. 6], so that $\Lambda_{24} = \Lambda_{24}$; the unsubscripted $\Lambda$ always denotes the minimal shell of $L$.

Every result below is a lower bound for $K$. Where the symbol $K(k)$ occurs inside a construction, the inequality proved holds with any lower bound for $K(k)$ substituted. The numerical values quoted use the best known one.

**Lemma 3.1** (the shell is a kissing configuration)**.** $\{\hat{v} : v \in \Lambda\}$ *is a kissing configuration in* $\mathbb{R}^n$*; hence* $K(n) \geq N$*.*

*Proof.* For distinct $v, w \in \Lambda$ the difference $v - w$ is a non-zero lattice vector, so $|v-w|^2 = 2\mu - 2\langle v, w\rangle \geq \mu$, giving $\langle v, w\rangle \leq \mu/2$ and hence $\langle \hat{v}, \hat{w}\rangle \leq \frac{1}{2}$. □

$L$ is *extremal* if its minimum is the largest permitted by the theory of modular forms, namely $\mu = 2\lfloor n/24\rfloor + 2$.

**Proposition 3.2** (extremal theta series)**.** *The theta series of an extremal even unimodular lattice of dimension* $n$ *is the unique modular form of weight* $n/2$ *for* $\mathrm{SL}_2(\mathbb{Z})$ *with constant term* 1 *whose coefficients of* $q^1, \dots, q^{\lfloor n/24\rfloor}$ *vanish. In particular* $(\mu, N)$ *depends only on* $n$*:*

| $n$ | 8 | 24 | 48 | 72 |
|---|---|---|---|---|
| $\mu$ | 2 | 4 | 6 | 8 |
| $N$ | 240 | 196 560 | 52 416 000 | 6 218 175 600. |

Four extremal lattices are known in dimension 48: $P_{48p}$ and $P_{48q}$ [13, Ch. 7], together with $P_{48m}$ and $P_{48n}$. Nebe constructed the fourth of them [39]; the catalogue [40] records all four. One extremal lattice is known in dimension 72, the lattice $\Gamma_{72}$ of Nebe [38]. Nothing in §B distinguishes the four, the bounds there being read off the Gram matrix and the one-point distribution, and we write $P_{48}$ for any of them; the searches on explicit coordinates in §C.4 are carried out in $P_{48p}$.

**Proposition 3.3** (Venkov [49])**.** *If* $L$ *is extremal even unimodular of dimension* $n \equiv 0 \pmod{24}$*, then* $\{\hat{v} : v \in \Lambda\}$ *is a spherical* 11*-design: every polynomial of degree at most* 11 *has the same average over* $\Lambda$ *as over the sphere. For* $n = 8$ *the shell is a* 7*-design.*

Integrality bounds the inner products in the shell.

**Lemma 3.4.** *For $v, w \in \Lambda$ the inner product $\langle v, w\rangle$ is an integer, and $|\langle v, w\rangle| \le \mu/2$ unless $w = \pm v$, in which case $\langle v, w\rangle = \pm\mu$.*

*Proof.* Integrality because $L$ is integral. Both $|v - w|^2 = 2\mu - 2\langle v, w\rangle$ and $|v + w|^2 = 2\mu + 2\langle v, w\rangle$ are norms of lattice vectors, so each is either 0 or at least $\mu$. If neither vanishes, the two bounds give $|\langle v, w\rangle| \le \mu/2$; and $|v \mp w|^2 = 0$ exactly when $w = \pm v$, which gives $\langle v, w\rangle = \pm\mu$. □

Thus for $P_{48}$ the inner products lie in $\{0, \pm1, \pm2, \pm3\} \cup \{\pm6\}$, and for $\Gamma_{72}$ in $\{0, \pm1, \pm2, \pm3, \pm4\} \cup \{\pm8\}$.

## 4. Current results

Table 1 gives the results dimension by dimension. The "recorded comparison" is the baseline available when the corresponding construction was recorded: we use Cohn's table when it contains that dimension, and otherwise the dated comparison floor defined in Section 1.2, obtained from a published configuration by an orthogonal direct sum or monotonicity. The "construction" column identifies the mathematical recipe that produces the entry in the "this work" column (for example, a lattice layer, a code layer or a cross-section) and points to its full definition and proof in the appendix. The related-work discussion records later improvements to those baselines.

## 5. The results

The table below is the current state-of-the-art summary. It records the current value by dimension and names the construction family; full construction descriptions and proofs are in the appendix.

Table 1: Current SOTA lower bounds in the reviewed sources. The reviewed set contains 65 certified claims; the superseded claims in dimensions 25, 29–31, 38 and 39 are retained in Appendix F.

| $n$ | recorded comparison | this work | construction |
|---|---|---|---|
| 18 | 7 654 | 8 358 | bent-coset hexagon over odd $\mathrm{BW}_{16}$ (§E) |
| 26 | 199 868 | 200 918 | shared midpoint replacements (§A.4) |
| 27 | 201 589 | 204 636 | shared midpoint replacements (§A.4) |
| 28 | 205 172 | 205 232 | joint Leech layers (§A.5) |
| 36 | 484 760 | 501 633 | Coxeter–Todd phase code (§A.1) |
| 37 | 500 008 | 505 359 | Coxeter–Todd phase code with latitude layers (§A.1) |
| 41 | 1 324 472 | 1 460 036 | $P_{48n}$ heads (§A.2) |
| 42 | 1 792 386 | 1 991 445 | $P_{48n}$ heads (§A.2) |
| 43 | 2 553 792 | 2 919 743 | $P_{48n}$ heads (§A.2) |
| 44 | 4 153 600 | 4 939 281 | $P_{48n}$ heads (§A.2) |
| 45 | 7 380 090 | 7 655 056 | $P_{48n}$ heads (§A.2) |
| 46 | 12 309 600 | 13 387 797 | $P_{48n}$ heads (§A.2) |
| 49 | 52 416 002 | 52 430 776 | layers over $P_{48}$ (§C.4) |

| $n$ | recorded comparison | this work | construction |
|---|---|---|---|
| 50 | 52 416 006 | 52 460 322 | layers over $P_{48}$ (§C.4) |
| 51 | 52 416 012 | 52 504 618 | layers over $P_{48}$ (§C.4) |
| 52 | 52 416 024 | 52 593 106 | layers over $P_{48}$ (§C.4) |
| 53 | 52 416 040 | 52 710 864 | layers over $P_{48}$ (§C.4) |
| 54 | 52 416 072 | 52 945 570 | layers over $P_{48}$ (§C.4) |
| 55 | 52 416 126 | 53 339 086 | layers over $P_{48}$ (§C.4) |
| 56 | 52 416 240 | 54 159 098 | layers over $P_{48}$ (§C.4) |
| 57 | 52 416 306 | 54 627 038 | layers over $P_{48}$ (§C.4) |
| 58 | 52 416 510 | 56 043 152 | layers over $P_{48}$ (§C.4) |
| 59 | 52 416 604 | 56 680 448 | layers over $P_{48}$ (§C.4) |
| 60 | 52 416 841 | 57 690 671 | layers over $P_{48}$ (§C.4) |
| 61 | 52 417 154 | 71 310 488 | binary code layers (§A.3) |
| 62 | 52 417 932 | 100 371 468 | binary code layers (§A.3) |
| 63 | 52 418 564 | 182 407 428 | binary code layers (§A.3) |
| 64 | 331 737 984 | 332 170 112 | signed-support code layer (§A.3) |
| 65 | 331 737 986 | 332 238 338 | signed-support code layer (§A.3) |
| 66 | 331 737 990 | 338 292 100 | binary code layers (§A.3) |
| 67 | 331 737 996 | 351 400 326 | binary code layers (§A.3) |
| 68 | 331 737 984 | 371 521 432 | binary code layers (§A.3) |
| 69 | 331 737 984 | 627 822 180 | $\Gamma_{72}$ cross-section (§B.8) |
| 70 | 331 737 984 | 1 249 778 250 | $\Gamma_{72}$ cross-section (§B.8) |
| 71 | 331 737 984 | 2 603 658 750 | $\Gamma_{72}$ cross-section (§B.8) |
| 73 | 6 218 175 602 | 6 218 191 606 | $\Gamma_{72}$ cosets and caps (§A.3) |
| 74 | 6 218 175 606 | 6 218 211 606 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 75 | 6 218 175 612 | 6 218 246 060 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 76 | 6 218 175 624 | 6 218 315 288 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 77 | 6 218 175 640 | 6 218 407 344 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 78 | 6 218 175 672 | 6 218 518 372 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 79 | 6 218 175 726 | 6 218 584 940 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 80 | 6 218 175 840 | 6 219 474 300 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 81 | 6 218 175 906 | 6 219 649 576 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 82 | 6 218 176 110 | 6 220 001 536 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 83 | 6 218 176 204 | 6 220 534 360 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 84 | 6 218 176 441 | 6 221 557 196 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 85 | 6 218 176 754 | 6 222 004 840 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 86 | 6 218 177 532 | 6 225 165 512 | nearest $\Gamma_{72}$ cosets (§A.3) |
| 87 | 6 218 178 164 | 6 225 838 030 | $\Gamma_{72}$ cap layers (§C.4) |
| 88 | 6 218 179 920 | 6 232 239 528 | $\Gamma_{72}$ cap layers (§C.4) |
| 89 | 6 218 181 330 | 6 235 525 036 | $\Gamma_{72}$ cap layers (§C.4) |
| 90 | 6 218 183 254 | 6 242 027 140 | $\Gamma_{72}$ cap layers (§C.4) |
| 91 | 6 218 187 548 | 6 252 202 550 | $\Gamma_{72}$ cap layers (§C.4) |
| 92 | 6 218 195 048 | 6 272 431 575 | $\Gamma_{72}$ cap layers (§C.4) |
| 93 | 6 218 205 368 | 6 301 636 084 | $\Gamma_{72}$ cap layers (§C.4) |
| 94 | 6 218 225 496 | 6 357 657 396 | $\Gamma_{72}$ cap layers (§C.4) |
| 95 | 6 218 268 750 | 7 091 452 684 | binary code layers (§A.3) |
| 96 | 6 218 372 160 | 13 289 406 656 | binary code layers (§A.3) |

The eight claims that have since been superseded remain useful as a record of the search and are documented in Appendix F.

## 6. Remarks

6.1. **The margin above dimension 48.** Among the dimensions at issue the tables have a single entry, at $n = 80$, where it agrees with the direct sum of §1.2. Everywhere else above dimension 48 the value a bound has to exceed is not a tabulated one but the largest a known construction gives. That construction is Edel–Rains–Sloane, evaluated in the dimension itself rather than inherited from a smaller one (Remark B.13). We therefore ran the chain of §D in every dimension of Table 1 above 48, with the best published code at each level, and asked of each bound how large the level-one constant-weight code $A(n, n_1, n_1)$ would have to be for the chain to overtake it. Divided by the best code available at those parameters, that is the factor by which the published constructions would have to be beaten.

No bound of Table 1 falls below the chain, and the factors are far from uniform. *Dimension* 68 *is the most exposed bound in this paper*: its margin over the chain is 8.9%, its level-one weight is 16, and 16 is the one weight at which Edel–Rains–Sloane built a strong code of their own, giving $A(64, 16, 16) \geq 30\,828$. A code with $A(68, 16, 16) \geq 45\,235$ would overturn Theorem B.12 in that dimension, a factor 1.47; for calibration, 30 828 beats the grid map of §D.1 at the same parameters by 1.88. No $A(68, 16, 16)$ has been published, so the exposure is to a better code and not to the literature as it stands. The next smallest factor is 4.06, in dimension 95; every other dimension exceeds 5. Dimensions 62, 63 and 96 are the chain, so the question does not arise there.

6.2. **The maximum independent set in $\Gamma_\gamma(\Lambda_{24})$.** Section C depends on the largest number of minimal lines of $\Lambda_{24}$ that are pairwise at $|\cos| \leq \frac{1}{4}$. The record is 248 lines [34], after 240 [9] and 244 [24]; the best upper bound is $\lfloor 4680/11 \rfloor = 425$ [9, §7]. By (4) one further line in every independent set contributes $2\sum_i(|Z_i| - 1)$, so raising the record by a single line gives $+2$, $+8$, $+16$, $+32$, $+52$, $+96$ and $+168$ in dimensions $25, \ldots, 31$ respectively.

The Delsarte linear programme, the Lovász theta function and the Hoffman ratio bound on the association scheme whose relations on the 98 280 lines are $|\langle u, u' \rangle| \in \{0, 1, 2\}$ all give 4680/11, so no two-point relaxation improves it. The three-point semidefinite bound of Bachoc and Vallentin [1] does not improve it at any truncation computed, the positivity constraints never being active. Closing the gap requires a four-point relaxation or an argument using the lattice structure. B. Lindow has since obtained 418 for lines of minimal vectors of $\Lambda_{24}$, by a three-point bound over the Terwilliger algebra of the automorphism group of $\Lambda_{24}$ with an exact rational dual (private communication, September 2026); for an arbitrary line system in $\mathbb{R}^{24}$ at $|\cos| \leq \frac{1}{4}$ the bound 425 stands.

6.3. **Verification.** The package [26] produces and checks every number above. It applies the following standards.

- *No floating-point arithmetic decides any bound of Table 1.* Every linear programming bound carries an exact rational dual certificate (Lemma B.6); the solver is used only to propose a dual vector. What decides the remaining bounds is an integer comparison. Two scans are too large to run in arbitrary-precision integers — the axis layer of dimension 95 compares $93\,150^2$ inner products — and are run in floating point on integer operands bounded so that every product and every partial sum is exactly representable. The bound is asserted before the scan; where it does not hold, the operand is split into two limbs at bit 31 and recombined exactly.
- *Every Gram matrix is proved realizable before use*, by Lemma B.7 or by exhibiting a $k$-tuple that carries it. The ones neither licensed nor exhibited are recorded and discarded.
- *Configurations small enough to write down are checked in coordinates.* The dimension-27 configuration is verified by two deliberately different arguments, one of which compares all $\binom{200540}{2} = 20\,108\,045\,530$ pairs; that configuration is now superseded (§F.4), and the two configurations of §F.4, in dimensions 26 and 27, are verified in exact arithmetic, the head–axis inequalities in $\mathbb{Q}(\sqrt{2},\sqrt{3})$. The four of §F.5 are verified the same way, in $\mathbb{Q}(\sqrt{2},\sqrt{3})$, and each of those checks includes that the four families of points have pairwise distinct $(|x|^2, |y|^2)$ — a statement about the point set rather than about any angle, and the one the count rests on.
- *The cross-section method is validated against every independently known value* (Table 2). The inequalities and the counting formula of the layered construction are re-derived from first principles in exact rational arithmetic, independently of the code that applies them.
- *Large intermediate data is regenerated by deterministic scripts* that re-verify every property the claims depend on.

## Appendix A. Current construction descriptions

The following sections give the construction descriptions and proofs behind the summary table. They are separated from the main results so that the main text can be read as a dimension-by-dimension account of the current bounds.

A.1. **Coxeter–Todd and $P_{48n}$ entries.** The dimension-36 and 37 constructions use three Coxeter–Todd quotient codes. Exact short-shell enumeration and phase-orbit angle checks certify 501 633 and 505 359, with two latitude layers in dimension 37.

A.2. **$P_{48n}$ heads.** The dimensions 41–46 use independently normalized $P_{48n}$ body families with rational heads. The package replays all integer angles and exact head ranks; the complete-family populations are inherited from the documented identities.

A.3. **Code layers and nearest cosets.** The code-layer packages give the current entries in dimensions 61–68 and 95–96; dimensions 64 and 65 use the signed-support parity augmentation. The nearest-coset and cap packages give the entries in dimensions 73–94 by exact group actions, repaired nearest subsets and integer checks.

In dimensions 95 and 96, the current support certificate has 24 561 subsets of size 24 in a $24 \times 4$ coordinate array. Each occupies at most twelve rows, and every distinct pair meets in at most twelve cells, as checked directly for all 301 609 080 pairs. The search allows different pairings in different rows, extending the uniform doubled-QR construction. Place a binary $[24, 14, 6]$ sign code, scaled by two, on each support. These points have squared norm 96 and mutual inner products at most 48. They are compatible with the existing dense layer and with the transversal layer, since a transversal takes only one cell from each row. The lower sparse layer has six entries of magnitude four, giving cross products at most $6 \cdot 4 \cdot 2 = 48$. On the central section $x_0 + \cdots + x_7 = 0$, reorder signs separately on each support. Intersection sizes $j = 0, 2, 4, 6, 8$ retain respectively 16384, 8192, 6144, 5120 and 8960 sign words; the last case uses a weight-eight dual word. These exact populations give $K(95) \geq 7\,091\,452\,684$ and $K(96) \geq 13\,289\,406\,656$. The independent certificate replay is `mixed_pair_certificate.py`.

In dimension 73, complete the half-coset to 6 272 representatives of squared norm 22, giving a lattice with 6 218 188 144 contacts. Retain its two layers $(x/2, \pm\sqrt{5/2})$. Select a class $C$ of 1731 minimal lines $u$ with $|\langle u, v\rangle| \leq 2$ inside $C$ and $|\langle u, x\rangle| \leq 4$ against every coset representative. Replace each equatorial pair $\pm u$ by the four points $(\pm\sqrt{3}u/2, \pm\sqrt{2})$. Their squared norms are eight. Within a direction the largest product is four, between distinct directions it is at most $7/2$, and against the coset layers it is at most $\sqrt{3} + \sqrt{5} < 4$. Retained equator products are at most $2\sqrt{3} < 4$. This adds $2|C|$ contacts and proves $K(73) \geq 6\,218\,191\,606$. Integer coordinates and all mixed products are checked by the package's `verify73.py`. The construction is non-lattice: two retained equatorial vectors sum to a removed $u$, and two caps sum to $\sqrt{3}u$, which cannot both belong to a discrete lattice.

A.4. **Shared midpoint replacements in dimensions 26 and 27.** Scale the Leech minimal shell to squared norm 32, so the kissing threshold is 16. If $u, v$ are minimal vectors with $\langle u, v\rangle = 16$, their midpoint $h = (u + v)/2$ has squared norm 24. Adjoin a tail $t$ of squared norm 8. For every retained equatorial point $z$ distinct from both endpoints,

$$\langle h, z\rangle = \tfrac{1}{2}(\langle u, z\rangle + \langle v, z\rangle) \leq 16.$$

Thus several midpoint heads can share the deletion of their endpoints.

For a local replacement, take mutually orthogonal $r, p, q$ of squared norm 16, with $r \pm p, r \pm q$ in the Leech shell, and orthogonal tail vectors $a, b$ of squared norm 8. Delete those four endpoints and insert

$$\big(r \pm (p + q)/2,\ \pm a\big), \qquad \big(r \pm (p - q)/2,\ \pm b\big),$$

with independent signs. These eight points form a translated cross-polytope and meet the threshold, gaining four points from four deletions. Compatible groups of these replacements give 8704 inserted points after 4352 deletions in dimension 26, and 16128 after 8064 deletions in dimension 27. Add a pure-tail hexagon or cuboctahedron, respectively. Each pure tail has squared norm 32, so Cauchy–Schwarz bounds its product with an inserted tail by $\sqrt{32 \cdot 8} = 16$. Consequently

$$K(26) \geq 200\,918 = 196560 - 4352 + 8704 + 6, \qquad K(27) \geq 204\,636 = 196560 - 8064 + 16128 + 12.$$

These exceed the 199868 and 201589 benchmarks of Takhanov–Yun [48] by 1050 and 3047. The portable package `dim26-27-midpoint-replacements` rebuilds the Golay Leech shell, verifies deleted-coordinate membership and every midpoint identity, and checks all inserted/inserted products with integer arithmetic. It uses no search caches. Retained equator pairs use the standard Leech kissing property; the convexity and pure-tail bounds above cover the remaining pairs. No optimum for this construction family is asserted.

A.5. **The current 28-dimensional lift.** The joint Leech-layer package selects even heads and auxiliary layers over the odd Leech blocks and certifies $K(28) \geq 205\,232$ in exact arithmetic.

## Appendix B. Cross-sections of extremal lattices

B.1. **Cross-sections and their cells.**

**Lemma B.1.** *Let $u_1, \dots, u_k \in \Lambda$ be linearly independent and put*

$$S = \{\, v \in \Lambda : \langle u_i, v \rangle = 0 \text{ for } i = 1, \dots, k \,\}.$$

*Then $S$ lies in a subspace of dimension $n - k$ and $K(n-k) \geq |S|$. We call $S$ the* cross-section *of $\Lambda$ by $u_1, \dots, u_k$.*

*Proof.* $S$ lies in $\operatorname{span}(u_1, \dots, u_k)^\perp$. All its members have norm $\mu$; Lemma 3.1 applies verbatim. □

It remains to count $S$. By Lemma 3.4 the vector $(\langle u_1, v \rangle, \dots, \langle u_k, v \rangle)$ attached to $v \in \Lambda$ lies in $\mathcal{I}^k$, where $\mathcal{I} = \{0, \pm 1, \dots, \pm\mu/2\} \cup \{\pm\mu\}$. We call an element $c = (c_1, \dots, c_k)$ of $\mathcal{I}^k$ a *cell* and write

$$n[c] \;=\; n[c_1, \dots, c_k] \;=\; \#\{\, v \in \Lambda : \langle u_i, v \rangle = c_i \text{ for } i = 1, \dots, k \,\}$$

for the number of minimal vectors it holds, the *count* of $c$. The notation leaves $u_1, \dots, u_k$ implicit, one cross-section being under discussion at a time; at $k = 1$ its argument is a single integer. Thus $|S| = n[0, \dots, 0]$, the counts of the $|\mathcal{I}|^k$ cells summing to $N$. The family $\big(n[c]\big)_{c \in \mathcal{I}^k}$ is the *$k$-point distribution* of $u_1, \dots, u_k$.

B.2. **Moments from the design property.**

**Lemma B.2.** *Let $t$ be the strength of the spherical design $\{\hat{v} : v \in \Lambda\}$ of Proposition 3.3, so $t = 11$ when $24 \mid n$ and $t = 7$ for $E_8$. For every exponent vector $p = (p_1, \dots, p_k)$ of non-negative integers with $\sum p_i \le t$ and $\sum p_i$ even,*

$$\sum_c n[c] \prod_{i=1}^{k} \Big(\frac{c_i}{\mu}\Big)^{p_i} \;=\; \frac{N \cdot \mathbb{E}\big[\prod_i \langle \hat{u}_i, \zeta\rangle^{p_i}\big]}{n(n+2)\cdots(n+\sum p_i - 2)}, \tag{1}$$

*where $\zeta$ is a standard Gaussian vector in $\mathbb{R}^n$ and the numerator is evaluated from the normalised Gram matrix $\big(\langle \hat{u}_i, \hat{u}_j\rangle\big)$ by Wick's theorem.*

*Proof.* The monomial $x \mapsto \prod_i \langle \hat{u}_i, x\rangle^{p_i}$ has degree $\sum p_i \le 11$, so by Proposition 3.3 its sum over the rescaled shell equals $N$ times its average over the sphere. Writing $x = \zeta/|\zeta|$ and using the independence of $|\zeta|$ and $\zeta/|\zeta|$ gives the normalisation $\mathbb{E}|\zeta|^{2m} = n(n+2)\cdots(n+2m-2)$. The Gaussian moment is a sum over perfect matchings, each pair contributing a Gram entry. Odd total degree vanishes because $\Lambda = -\Lambda$. Finally $\langle \hat{u}_i, \hat{v}\rangle = \langle u_i, v\rangle/\mu$ since all vectors have norm $\mu$. □

*Remark* B.3 (mixed exponents). Imposing (1) only for those $p$ with every $p_i$ even gives a valid but weaker programme, whose value depends on the presentation of the Gram matrix. At $k = 3$ the full family has 161 equations and the even-exponent family 56. Write a $3 \times 3$ Gram matrix of minimal vectors by its off-diagonal entries. Write $3A_3$ for three times the Gram matrix of the root lattice $A_3$. The matrix $G = (3, -3, 0)$ satisfies $U^{\mathsf{T}} G U = 3A_3$ for an integer matrix $U$ with $\det U = -1$; thus $G$ and $3A_3$ describe the same configuration. With even exponents alone the two presentations give 6 462 480 and 5 194 969; with the full family both give 6 687 320.

B.3. **Constraints from the lattice structure.** Proposition 3.3 uses only that $\Lambda$ is a spherical 11-design. The next two lemmas sharpen the bound by using that $\Lambda$ is the minimal shell of a lattice.

**Lemma B.4** (translations)**.** *Suppose $w = \sum a_i u_i$ is a lattice vector with $|w|^2 = \mu$, and let $\tau_l = \langle u_l, w\rangle$. If $\langle w, v\rangle = \mu/2$ for $v \in \Lambda$, then $w - v \in \Lambda$ and $v \mapsto w - v$ is an involution. Consequently*

$$n[c] = n[c'], \qquad c'_l = \tau_l - c_l,$$

*for every cell $c$ with $\sum_i a_i c_i = \mu/2$. If $c' \notin \mathcal{I}^k$ the constraint becomes $n[c] = 0$.*

*Proof.* $|w - v|^2 = 2\mu - 2\langle w, v\rangle = \mu$, so $w - v$ is minimal, with $\langle u_l, w - v\rangle = \tau_l - c_l$. The hypothesis $\langle w, v\rangle = \sum a_i c_i = \mu/2$ depends only on the cell; the involution therefore maps the cell $c$ onto the cell $c'$. □

**Lemma B.5** (lattice range)**.** *For every lattice vector $w = \sum a_i u_i$ in the span, with $|w|^2 = \nu > 0$, and every $v \in \Lambda$,*

$$|\langle v, w\rangle| \le \nu/2 \qquad \text{unless } v = \pm w \text{ (possible only if } \nu = \mu).$$

*A cell violating the inequality for some such $w$ is empty, unless it is the cell of $\pm w$ itself, which then holds exactly one vector.*

*Proof.* $|v \mp w|^2 = \mu + \nu \mp 2\langle v, w\rangle$ is a lattice norm, hence 0 or at least $\mu$. □

Lemma B.5 is strictly stronger than testing positive semidefiniteness of the Gram matrix of $(u_1, \dots, u_k, v)$: for a 60° pair, $u_1 - u_2$ is itself minimal, which forces $|c_1 - c_2| \le \mu/2$ outside the two cells of $\pm(u_1 - u_2)$ and so excludes ten of the twelve cells that semidefiniteness permits. Those ten cells are the zeros of [44, Lemma 4.2].

Two further exact constraints are available. The first is the antipodal symmetry $n[c] = n[-c]$. The second is the set of one-point marginals, which are Lemma B.2 at $k = 1$; §B.6 solves them in closed form.

B.4. **The linear programme and its dual certificate.** Call the linear programme of this subsection, built from $k$ fixed vectors, the *$k$-point programme*. Its equality constraints form the *$k$-point system*.

For a cell $c$ let $G(c)$ be the Gram matrix of $(u_1, \dots, u_k, v)$ for a vector $v$ of norm $\mu$ with $\langle u_i, v\rangle = c_i$ for every $i$. Its entries are determined by $c$ and by the Gram matrix of $u_1, \dots, u_k$, whether or not such a $v$ lies in $\Lambda$.

Collect the constraints as $Ax = b$, $0 \le x \le h$, with $x$, $h$ and $s$ below all indexed by cells. Set $h_c = 1$ when $\det G(c) = 0$ and $h_c = +\infty$ otherwise. A cell with $\det G(c) < 0$ is the profile of no real vector at all and is forced to zero, and a cell that is the profile of a minimal lattice vector of $\operatorname{span}(u_1, \dots, u_k)$ is forced to one. The first case is legitimate: a vector $v \in \Lambda$ counted by such a cell lies in $\operatorname{span}(u_1, \dots, u_k)$ and is determined there by $c$; at most one such $v$ therefore exists. Let $e_0$ be the indicator of the all-zero cell. The true distribution is feasible, whence

$$|S| \;\ge\; \min\{\, e_0^{\mathsf T} x : Ax = b,\; 0 \le x \le h \,\}.$$

**Lemma B.6** (weak duality)**.** *For any rational vector $y$, put $s = e_0 - A^{\mathsf T} y$. If $s_c \ge 0$ for every cell $c$ with $h_c = +\infty$, then*

$$\min\{\, e_0^{\mathsf T} x : Ax = b,\; 0 \le x \le h \,\} \;\ge\; b^{\mathsf T} y - \sum_{c:\, h_c < \infty} h_c \max(0, -s_c).$$

*Proof.* For feasible $x$ we have $e_0^{\mathsf T} x = y^{\mathsf T} A x + s^{\mathsf T} x = b^{\mathsf T} y + s^{\mathsf T} x$, and $s^{\mathsf T} x \ge -\sum_{h_c < \infty} h_c \max(0, -s_c)$ because $0 \le x_c \le h_c$. □

A floating-point solver is used only to propose a candidate $y$. That $y$ is rounded to rationals and repaired into exact dual feasibility by decreasing the coordinate belonging to the all-ones moment row, whose coefficients are all positive. The right-hand side of Lemma B.6 is then evaluated in exact rational arithmetic and rounded to an integer downwards. An imprecise $y$ weakens the bound and cannot invalidate it. Applying the same lemma to $-e_0$ certifies an upper bound on $|S|$, rounded upwards. A certified minimum may therefore fall one below the true optimum, and a certified maximum one above it.

B.5. **Realizability.** Call a symmetric $k \times k$ integer matrix $G$ with every diagonal entry equal to $\mu$ *realizable in* $L$ if some $u_1, \dots, u_k \in \Lambda$ have Gram matrix $G$. A bound computed for $G$ is vacuous unless $G$ is realizable in $L$.

**Lemma B.7.** *Let $u_1, \dots, u_{k-1} \in \Lambda$ have Gram matrix $G'$, and let $c \in \mathcal{I}^{k-1}$. Suppose the minimum of $n[c]$ over the $(k-1)$-point programme is strictly positive. Then some $v \in \Lambda$ satisfies $\langle u_i, v\rangle = c_i$ for $i = 1, \dots, k-1$, and the matrix obtained by bordering $G'$ with the row $c$ and the diagonal entry $\mu$ is realizable in $L$.*

*Proof.* The true distribution is feasible for the $(k-1)$-point programme; its value at the cell $c$ is therefore at least the minimum, hence positive. Take $u_k = v$. □

Lemma B.7 is the weaker of the two routes to realizability. A sweep over all Gram matrices produces larger apparent bounds whose bordering cells have minimum 0, about which the lemma says nothing; among three-point Gram matrices in dimension 69 it licenses only the pairwise orthogonal one, yielding 149 415 783, below the value 331 737 984 of §1.2. Realizability is, however, a property of the lattice and not of the programme. Where the lattice is available in coordinates a $k$-tuple with a prescribed Gram matrix can simply be *exhibited*, as it is for $\Gamma_{72}$: dimensions 68 and 69 of Table 1 are obtained this way.

Which Gram matrix to exhibit is then guided by a measurement. Across the Gram matrices tried here the bound falls as $\det G$ rises — at $k = 3$ from 627 822 180 at $\det G = 256$ through 527 597 644 at 352 to 149 415 783 at 512. At a fixed determinant the presentation makes no difference: at $k = 3$ the sixteen realizable Gram matrices of determinant 256 are sixteen bases of one scaled $A_3$, and all sixteen certify 627 822 180. Taking the determinant as low as it will go, $u_1, \dots, u_k$ generate a sublattice $M$ of rank $k$ with $\min M \geq \mu$ and $\det M = \det G$, so Hermite's inequality gives $\det M \geq (\mu/\gamma_k)^k$, where $\gamma_k$ is Hermite's constant in rank $k$, with equality exactly for the densest rank-$k$ lattice. For $\Gamma_{72}$, where $\mu = 8$, that is 48 at $k = 2$, 256 at $k = 3$ and 1024 at $k = 4$, attained by $A_2$, $A_3$ and $D_4$. Those are the three Gram matrices used in dimensions 70, 69 and 68; each is exhibited inside $\Gamma_{72}$. Dimension 69 gains 100 224 536 over the Gram $(4, -2, 0)$, whose determinant is 352.

B.6. **The one-point distribution.** At $k = 1$ the system of Lemma B.2 has $\mu/2 + 1$ unknowns and six equations, one for each even degree up to 10; the shell of $E_8$ is a 7-design, which gives four equations there. The system is over-determined; the surplus equations serve as a consistency check.

**Proposition B.8.** *For $u \in \Lambda$ and $c = 0, \dots, \mu/2$ let $n[c]$ be the number of $v \in \Lambda$ with $\langle u, v\rangle = c$. Then*

| *lattice* | $n[0]$ | $n[1]$ | $n[2]$ | $n[3]$ | $n[4]$ |
|---|---|---|---|---|---|
| $E_8$ | 126 | 56 | | | |
| $\Lambda_{24}$ | 93 150 | 47 104 | 4 600 | | |
| $P_{48}$ | 23 766 960 | 12 608 784 | 1 678 887 | 36 848 | |
| $\Gamma_{72}$ | 2 603 658 750 | 1 512 243 200 | 280 928 256 | 13 959 168 | 127 800 |

*with each non-zero $c$ attained by $n[c]$ vectors of each sign. The surplus equations hold in every case.*

The first two rows are the root system $E_7$ and the laminated lattice $\Lambda_{23}$, whose kissing numbers are known independently. The $\Gamma_{72}$ row and Lemma B.1 give Theorem B.11; the $P_{48}$ row recovers a published value (§B.7).

B.7. **Dimensions 46 and 47.** Both values below are in the literature (§1.5). We derive them here to test the programme against a published value.

**Proposition B.9.** *The cross-section of $P_{48}$ by one minimal vector has exactly* 23 766 960 *elements, and the cross-section by two minimal vectors at* 60° *has exactly* 12 309 600. *Hence $K(47) \geq$ 23 766 960 and $K(46) \geq$ 12 309 600. Both values are known, the first from* [3] *and the second from* [46].

*Proof.* For $k = 1$, Proposition B.8 gives $n[0] =$ 23 766 960 and the system determines it uniquely; Lemma B.1 finishes.

For $k = 2$ take $\langle u_1, u_2 \rangle = 3$, a 60° pair. Such a pair exists: the one-point programme has value $n[3] = 36\,848 > 0$ at that cell, so Lemma B.7 applies. Running the programme of Lemmas B.2–B.6 on the all-zero cell, first minimising and then maximising, returns the same certified value 12 309 600 in both directions. Since the true size lies between the two, it equals 12 309 600. □

For the orthogonal pair the programme gives an interval,

$$10\,659\,120 \;\leq\; |S| \;\leq\; 10\,697\,520,$$

and no bound depending only on the Gram matrix can be exact. In the notation of Remark B.10, row 20 of [44, Table 3] divided by $a((3,3,0))$ equals $458\,386\,320/43$, which is not an integer. That quotient is an exact average over ordered orthogonal pairs, so the size varies from pair to pair. The width of the interval reflects that variation.

*Remark* B.10 (modular forms). The dimension-46 value also follows from modular forms, with no linear programming. Write $a(T)$ for the Siegel Fourier coefficient of the theta series of $P_{48}$ at the half-integral form $T$, which counts the tuples of lattice vectors whose Gram matrix is $2T$; here $T$ is listed as its diagonal entries followed by the off-diagonal entries of $2T$, so that minimal vectors give diagonal 3 and pairwise entries $\langle u_i, u_j \rangle$. The

degree-3 coefficient at a 60° pair together with a vector orthogonal to both, over the degree-2 coefficient at a 60° pair, is

$$\frac{a\big((3,3,3,3,0,0)\big)}{a\big((3,3,3)\big)} = \frac{23\,775\,066\,324\,172\,800\,000}{1\,931\,424\,768\,000} = 12\,309\,600$$

exactly, with no remainder; the degree-3 coefficients are [44, Table 3]. Sun–Wang [46] obtain the same value by direct enumeration.

B.8. **Dimensions 68 to 71.**

**Theorem B.11.** *$K(71) \geq 2\,603\,658\,750$ and $K(70) \geq 1\,249\,778\,250$. The first is the exact size of that cross-section of $\Gamma_{72}$.*

*Proof.* Since $72 \equiv 0 \pmod{24}$, the shell of $\Gamma_{72}$ is an 11-design by Proposition 3.3. For a fixed $u \in \Lambda$, Lemma 3.4 allows $\langle u, v\rangle = c$ with $|c| \leq 4$, together with $c = \pm 8$ for $v = \pm u$: five unknowns against six moment equations. The system is over-determined, solves in non-negative integers, and the surplus equation holds (Proposition B.8). Then $\Gamma_{72} \cap u^{\perp}$ is a 71-dimensional lattice of minimum 8 whose minimal vectors are exactly the $n[0]$ vectors orthogonal to $u$; Lemma B.1 gives the first bound.

For $k = 2$ take $\langle u_1, u_2\rangle = 4$. Such a pair exists because $n[4] = 127\,800 > 0$. The two-point programme certifies 1 249 778 250. Sweeping all five admissible inner products $\langle u_1, u_2\rangle = 0, 1, 2, 3, 4$ gives, in that order, 1 073 551 500, 1 089 616 500, 1 117 183 200, 1 167 565 500 and 1 249 778 250; the 60° pair is the best of them. □

Unlike dimension 46, the minimum and maximum do not meet here: the interval is [1 249 778 250, 1 250 506 441], of which only the lower end is claimed.

At $k = 3$ and $k = 4$ the moment system no longer forces the bound and the Gram matrix has to be chosen; §B.5 takes one at the Hermite floor. That choice is guided by a measurement, but the bound does not rest on it: each certificate is valid for the Gram matrix it is computed from, and that matrix is exhibited.

**Theorem B.12.** *$K(69) \geq 627\,822\,180$ and $K(68) \geq 361\,275\,480$.*

The 68-dimensional value in this theorem is retained as a cross-section validation. The current state-of-the-art entry is the code-layer value $K(68) \geq 371\,521\,432$ in Table 1; the proof below concerns the cross-section programme and its independently checked lower value.

*Proof sketch.* For $k = 3$ take the Gram matrix with diagonal 8 and off-diagonal entries $-4, -4, 0$, that of $A_3$ scaled by 4, of determinant 256. For $k = 4$ take that of $D_4$ scaled by 4, of determinant 1024. By §B.5 these are the least determinants a triple and a quadruple of minimal vectors of $\Gamma_{72}$ can have. Both are realizable by exhibition. Four minimal vectors of $\Gamma_{72}$ with the second Gram matrix are written out in [26]; the rank-4 sublattice they generate has 24 minimal vectors, among which a triple with the

first is displayed. The three-point and four-point programmes then certify 627 822 180 and 361 275 480; Lemma B.1 finishes. □

What a numerically proposed dual can be repaired into depends on the basis; the programme's optimum does not. Of the 104 determinant-1024 Gram matrices realised by quadruples of those 24 minimal vectors, ninety-eight yield a repaired certificate, running from 358 774 871 to 361 275 383, a spread of 0.7%; the solver does not finish the other six. Solving for the dual *vertex* instead removes it. Substituting out the constraints that are identifications — $n[c] = n[-c]$, and the translations that read $n[c] = n[c']$, $n[c] = 0$ or $n[c] = 1$ — is exact and takes the $k = 4$ programme from 3473 rows and 1681 columns to 617 and 520; the active system is then solved in exact rationals and $A^{\mathsf{T}}y \le c$ verified over every column. The optimum is 361 275 480, and the numeric primal attains it, so no dual improves on it.

The chain ends at $k = 4$. The certified values fall by factors 0.480, 0.502 and 0.575 as $k$ runs from 1 to 4; reaching 331 737 984 at $k = 5$ would need a factor 0.92, and the Hermite floor there is 4096. We have not computed $k = 5$, where the cell enumeration is $11^k$; the trend is evidence that it would fall short, not a proof. Dimensions 65 through 67 are left to the current code-layer packages; this cross-section calculation makes no claim in them.

*Remark* B.13 (the Edel–Rains–Sloane value). No table has an entry in dimensions 65–71, so the value to improve on is 331 737 984, inherited from dimension 64 [19]. The chain of §D reproduces both of their large published totals:

$$2^{28} + 30828 \cdot 2048 + 10416 \cdot 16 + 128 = 331\,737\,984,$$

$$2^{30} + 143780 \cdot 2048 + 20540 \cdot 16 + 160 = 1\,368\,532\,064.$$

An inherited value is not the value the construction takes in the dimension that inherits it. The difference matters in dimension 68, where the margin is smallest. There the chain is $(64, 16, 4, 1)$, not $(68, \dots)$: the level-zero sign code has length 64, the best available at any length up to 68, while the support codes spend all 68 coordinates. It gives 331 771 800 from $A(68, 16, 16) \ge A(64, 16, 16) \ge 30\,828$ and $A(68, 4, 4) = \binom{68}{3}/4 = 12\,529$. The second of these is exact, since $68 \equiv 2 \pmod 6$ admits a Steiner system $S(3, 4, 68)$. Theorem B.12 exceeds that by a factor 1.089, the narrowest margin over the chain of any bound here not derived from it; §6.1 measures the others. Evaluated at $n = 70$ and $n = 71$, granting it the dimension-80 constant-weight code size it has no claim to at those lengths, the chain gives at most 563 125 646, which Theorem B.11 exceeds by factors 2.2 and 4.6.

B.9. **Validation against known cross-sections.** An invalid constraint inflates the minimum of the programme and makes the lower bound false. We therefore ran the programme on every cross-section whose size is known independently and checked that the known size lies inside the interval the programme produces. Table 2 reports the outcome.

TABLE 2. Every independently known cross-section, against the interval the method produces for it. Sources of truth: classical values for the laminated lattices; direct enumeration from coordinates; and the published computations of [3, 44, 46]. The orthogonal Gram matrices over $\Lambda_{24}$ at $k = 3$ and $k = 4$ admit embeddings with different counts: at $k = 3$ both are given, at $k = 4$ (†) one of several.

| lattice | configuration | minimum | **truth** | maximum |
|---|---|---|---|---|
| $E_8$ | $k = 1$ | 126 | **126** | 126 |
| $E_8$ | $k = 2$ orthogonal | 60 | **60** | 60 |
| $E_8$ | $k = 2$ at $60°$ | 72 | **72** | 72 |
| $\Lambda_{24}$ | $k = 1$ | 93 150 | **93 150** | 93 150 |
| $\Lambda_{24}$ | $k = 2$ orthogonal | 43 164 | **43 164** | 43 164 |
| $\Lambda_{24}$ | $k = 2$ at $\cos\frac{1}{4}$ | 44 550 | **44 550** | 44 550 |
| $\Lambda_{24}$ | $k = 2$ at $60°$ | 49 896 | **49 896** | 49 896 |
| $\Lambda_{24}$ | $k = 3$ orthogonal | 19 530 | **19 530**, **19 962** | 19 962 |
| $\Lambda_{24}$ | $k = 3$, $2A_3$ | 27 720 | **27 720** | 27 720 |
| $\Lambda_{24}$ | $k = 4$ orthogonal | 7 464 | **8 616**† | 17 400 |
| $\Lambda_{24}$ | $k = 4$, $2A_4$ | 15 540 | **15 540** | 15 540 |
| $\Lambda_{24}$ | $k = 4$, $2D_4$ | 17 400 | **17 400** | 17 400 |
| $P_{48}$ | $k = 1$ | 23 766 960 | **23 766 960** | 23 766 960 |
| $P_{48}$ | $k = 2$ at $60°$ | 12 309 600 | **12 309 600** | 12 309 600 |
| $P_{48}$ | $k = 3$, $3A_3$ | 6 687 320 | **6 687 648** | 6 702 736 |
| $P_{48}$ | $k = 3$, $3D_3$ | 6 687 320 | **6 702 080** | 6 702 736 |
| $P_{48}$ | $k = 4$, $3D_4$ | 4 147 967 | **4 153 600** | 4 165 632 |
| $\Gamma_{72}$ | $k = 1$ | 2 603 658 750 | **2 603 658 750** | 2 603 658 750 |

Every known size lies inside its interval.

Dimension 68 is a cross-section at $k = 4$, and on the Leech lattice the truth is available there. The programme is tight at the two Gram matrices of least determinant: for $2D_4$, which attains the Hermite floor 64, and for $2A_4$, of determinant 80, the minimum and the maximum coincide with the count. The orthogonal quadruple, of determinant 256 against $D_4$'s 64, shows that this is a property of the good Gram matrices rather than of the programme; at that Gram matrix no programme reading the Gram matrix alone could be exact, the count not being a function of it.

The size of a cross-section is not a function of the Gram matrix. The rows $3A_3$ and $3D_3$ carry one and the same Gram matrix — $A_3$ and $D_3$ are the same abstract lattice, so Sun–Wang's two dimension-45 complements have a single Gram matrix between them — so the programme returns one interval for both, yet their sizes differ: 6 687 648 and 6 702 080, at very nearly its two ends. Over $\Lambda_{24}$ the effect can be exhibited outright: in the coordinates in

which a minimal vector has shape $(\pm 4^2, 0^{22})$, the triples

$$4e_1{+}4e_2, \quad 4e_3{+}4e_4, \quad 4e_5{+}4e_6 \qquad \text{and} \qquad 4e_1{+}4e_2, \quad 4e_1{-}4e_2, \quad 4e_3{+}4e_4$$

are both pairwise orthogonal with every norm equal to 32, the minimum in these coordinates, so they share one Gram matrix $32I_3$, and their cross-sections have 19 530 and 19 962 elements — exactly the two ends of the certified interval. Orthogonal quadruples behave the same way, realising 8 616, 8 760 and 9 192 among others. A programme reading the Gram matrix alone can therefore only bracket such a row, and the truth column records one embedding in each of these two.

The rows with $k = 3$ also recover a theorem of Ozeki. The three-point system for a $60^\circ$ triple has affine solution space of dimension 1, along $n[0,0,0] + 164\,n[3,3,3] = 6\,732\,912$. Ozeki's range $184 \le n[3,3,3] \le 278$ [44, Theorem 6.3] therefore corresponds to $6\,687\,320 \le n[0,0,0] \le 6\,702\,736$, which is exactly the interval certified in Table 2; the outward rounding of §B.4 costs nothing at either end. Ozeki obtains the range from modular forms; the programme obtains it with an exact rational certificate.

## Appendix C. The layered construction

C.1. **The construction.** Fix a lattice $L$ as in §3 and an auxiliary dimension $k$, and work in $\mathbb{R}^n \oplus \mathbb{R}^k$. Choose a level $t \in (0,1)$ and write $a = \sqrt{t}$, $b = \sqrt{1-t}$.

A minimal *line* of $L$ is an antipodal pair $\pm u$ with $u \in \Lambda$. Set

$$\gamma = \frac{\lfloor \mu/3 \rfloor}{\mu}, \tag{2}$$

the largest inner-product threshold available in $L$ that does not exceed $\frac{1}{3}$. Thus $\gamma = \frac{1}{4}$ for $\Lambda_{24}$ ($|\langle u, u' \rangle| \le 1$ of 4) and for $\Gamma_{72}$ ($\le 2$ of 8), while $\gamma = \frac{1}{3}$ for $P_{48}$ ($\le 2$ of 6). Let $\Gamma_\gamma(L)$ be the graph on the minimal lines of $L$ in which two lines are adjacent when the cosine of the angle between them exceeds $\gamma$ in absolute value.

Let $C_1, \dots, C_r$ be pairwise disjoint independent sets of $\Gamma_\gamma(L)$, and let $Z_i \subset S^{k-1}$ be a set of unit *directions* attached to $C_i$. The configuration is

$$\begin{array}{lll} \text{equator} & (\hat{v},\, 0) & v \in \Lambda,\ v \neq \pm u \text{ for any } u \in \bigcup_i C_i, \\ \text{caps} & (s\,a\,\hat{u},\ b\,z) & s \in \{\pm 1\},\ u \in C_i,\ z \in Z_i, \\ \text{poles} & (0,\, w) & w \in W \subset S^{k-1}. \end{array} \tag{3}$$

Every point is a unit vector. We write $Z = \bigcup_i Z_i$ for the set of all directions. Counting, with $|C_i|$ the number of lines in $C_i$,

$$\# \;=\; N - 2\sum_i |C_i| \;+\; 2\sum_i |C_i|\,|Z_i| \;+\; |W| \;=\; N + 2\sum_i |C_i|\big(|Z_i| - 1\big) + |W|. \tag{4}$$

**Proposition C.1.** *The configuration* (3) *is a kissing configuration in* $\mathbb{R}^{n+k}$ *if and only if*

$$t \le 1, \tag{5}$$

$$t\gamma + (1-t) \le \tfrac{1}{2}, \tag{6}$$

$$\langle z, z' \rangle \le \frac{\frac{1}{2} - t}{1-t} \quad \text{for distinct } z, z' \in Z_i, \tag{7}$$

$$\langle z, z' \rangle \le \tfrac{1}{2} \quad \text{for } z \in Z_i,\ z' \in Z_j,\ i \ne j, \tag{8}$$

$$b\,\langle z, w \rangle \le \tfrac{1}{2} \quad \text{for } z \in Z,\ w \in W, \tag{9}$$

$$\langle w, w' \rangle \le \tfrac{1}{2} \quad \text{for distinct } w, w' \in W. \tag{10}$$

*Proof.* There are nine kinds of pair. Five are immediate. Equator against equator is Lemma 3.1 and equator against pole is 0. Equator against cap gives $s\,a\langle \hat{v}, \hat{u} \rangle$; since $v \ne \pm u$ forces $|\langle \hat{v}, \hat{u} \rangle| \le \frac{1}{2}$, the worst case is $a/2 \le \frac{1}{2}$, which is (5). Cap against pole gives $b\langle z, w \rangle$, which is (9), and pole against pole is (10).

The remaining four are cap against cap, and we take them by how the two caps share a line. On one line and one direction the two signs give $-t + (1-t) = 1-2t$, which is at most $\frac{1}{2}$ because (6) rearranges to $t \ge 1/(2(1-\gamma)) \ge \frac{1}{2}$. On one line and distinct directions the value is $\pm t + (1-t)\langle z, z' \rangle$, worst at $s = s'$, which is (7). On distinct lines of one $C_i$ it is $\pm t\langle \hat{u}, \hat{u}' \rangle + (1-t)\langle z, z' \rangle$ with $|\langle \hat{u}, \hat{u}' \rangle| \le \gamma$; at $z = z'$ that is at most $t\gamma + (1-t)$, which is (6), and at $z \ne z'$ it is at most $t\gamma + (1-t)\langle z, z' \rangle \le t\gamma + \frac{1}{2} - t < \frac{1}{2}$ by (7), so this case is implied by the two before it. On lines from distinct $C_i$ and $C_j$ only $|\langle \hat{u}, \hat{u}' \rangle| \le \frac{1}{2}$ is available, by Lemma 3.1, and $t/2 + (1-t)\langle z, z' \rangle \le \frac{1}{2}$ is (8). □

C.2. **Pairs versus triples.** Constraints (6) and (7) oppose one another. Rearranged, (6) becomes

$$t \;\ge\; \frac{1}{2(1-\gamma)}, \tag{11}$$

so a larger $\gamma$ forces a larger level. Substituting the smallest admissible $t$ into (7) bounds the inner products among directions attached to one independent set.

**Corollary C.2.** *At* $t = 1/(2(1-\gamma))$ *the directions attached to one independent set satisfy* $\langle z, z' \rangle \le \gamma/(2\gamma - 1)$. *Hence*

$$\begin{aligned} \gamma = \tfrac{1}{4}\ (\Lambda_{24},\ \Gamma_{72}): &\quad t \ge \tfrac{2}{3},\quad \langle z, z' \rangle \le -\tfrac{1}{2},\quad |Z_i| \le 3, \\ \gamma = \tfrac{1}{3}\ (P_{48}): &\quad t \ge \tfrac{3}{4},\quad \langle z, z' \rangle \le -1,\quad |Z_i| \le 2. \end{aligned}$$

*Proof.* Substituting $t = 1/(2(1-\gamma))$ into $(\frac{1}{2} - t)/(1-t)$ gives $\gamma/(2\gamma - 1)$. If $m$ unit vectors are pairwise at $\langle z, z' \rangle \le c < 0$ then $0 \le |\sum z_i|^2 \le m + m(m-1)c$, so $m \le 1 - 1/c$. With $c = -\frac{1}{2}$ this is $m \le 3$ and with $c = -1$ it is $m \le 2$. □

Corollary C.2 forces the ceiling $\frac{1}{3}$ in (2): at a larger threshold the right-hand side of (7) falls below $-1$ and no independent set carries more than one

direction. The ceiling is active for $\Gamma_{72}$, where the largest threshold below $\frac{1}{2}$ that Lemma 3.4 leaves available is $\gamma = \frac{3}{8}$, that is $|\langle u, u'\rangle| \le 3$; the lemma itself permits $|\langle u, u'\rangle| \le 4$, but $\gamma = \frac{1}{2}$ makes (6) require $t \ge 1$. The threshold $\frac{3}{8}$ forces $t \ge \frac{4}{5}$, at which (7) requires $\langle z, z'\rangle \le -\frac{3}{2}$, which no two unit vectors satisfy.

Over $\Lambda_{24}$ and $\Gamma_{72}$ an independent set may therefore carry a *zero-sum triple* of directions: three unit vectors summing to zero and pairwise at 120°. Over $P_{48}$ it carries at most an antipodal pair. By (4) a triple contributes 4 points per line and a pair 2, doubling the gain; no independent set carries more than three directions.

Constraint (9) behaves differently at the two levels used below. At $t = \frac{3}{4}$ we have $b = \frac{1}{2}$, so (9) holds for every $z$ and $w$; $W$ may then be any kissing configuration of $\mathbb{R}^k$, giving $|W| = K(k)$. At $t = \frac{2}{3}$ we have $b = 1/\sqrt{3}$ and (9) requires $\langle z, w\rangle \le \sqrt{3}/2$; every pole then lies at least 30° from every cap direction. A rotated copy of the direction set is automatically a 60° code, so only the 30° separation is at issue, and the caps around $Z$ occupy little of the sphere once $k$ is large: at $k = 23$ a single 30° cap covers $2.3 \times 10^{-8}$ of $S^{22}$, so all 93 150 of them together cover at most $2.1 \times 10^{-3}$ and a Haar-random rotation is expected to lose at most 198 directions; the rotation used below loses 76, leaving 93 074. We therefore take $W$ to be a rotated copy of $Z$ with the points that fail (9) removed. The rotation is made rational by the Cayley transform $Q = (I - S)(I + S)^{-1}$, which is an isometry of the form whenever $CS$ is skew-symmetric, so (9) is checked in exact integers. It must be an isometry of $\mathbb{R}^k$ itself and not merely of the coordinates in which $Z$ is presented, which for some cross-sections are more numerous: $Z$ spans 21 dimensions of the 24 that carry $\Lambda_{21}$. Where they differ we take $S = \sum_t \big(z_t(Cz_t')^{\mathsf{T}} - z_t'(Cz_t)^{\mathsf{T}}\big)$ over pairs $z_t, z_t' \in Z$ with $\langle z_t, z_t'\rangle = 0$, all $2r$ of them pairwise orthogonal. Then $S$ is integral, $CS$ is skew, and $S$ annihilates the orthogonal complement of $Z$, so $Q$ fixes that complement pointwise and is an isometry of the span. On the plane of $z_t$ and $z_t'$, taken in the integral coordinates where $\langle z_t, z_t\rangle = m$, scaling $S$ by $1/\lambda$ makes $Q$ the rotation through $2\arctan(m/\lambda)$, and its denominator is $\lambda^2 + m^2$ for every plane at once — two to four digits, against the $2^{71}$ that a basis of the span would force. Where the direction set is realised over a ring in which we have not done this we take $W = \emptyset$.

C.3. **Calibration.** We first check (4) in dimensions where the answer is known. Take $L = \Lambda_{24}$, so that $\gamma = \frac{1}{4}$, and take every $C_i$ to be a maximum known independent set, of 248 lines. Take $Z$ to be the root system of rank $k$ normalised to the sphere, partitioned into zero-sum triples where its size permits and into antipodal pairs otherwise; take $|W| = K(k)$ poles. The level is the smallest admissible one, $t = \frac{2}{3}$, in every column but the first: at $k = 1$ the sphere $S^0$ contains no triple; §F.2 raises the level to $t = \frac{3}{4}$ there. Where $t = \frac{2}{3}$ we take the poles to satisfy the 30° separation of (9). Such a

set of $K(k)$ poles is exhibited here in every column. At $k = 2$ and $k = 4$ the poles are vectors of norm $2m$ in the same root lattice, where $|a \pm w|^2$ being a lattice norm forces the 30° separation and $K(k)$ of them are available. At $k = 3$ they are the axis layer of the dimension-27 configuration of §F.3, verified over all of its pairs. At $k = 5$, $k = 6$ and $k = 7$ they are a rotated copy of the root system, constructed as in §C.4 and checked in exact integers there. So no column of Table 3 assumes a pole set: each is exhibited and verified. Then (4) gives Table 3.

TABLE 3. The total (4) over $\Lambda_{24}$ with independent sets of 248 lines, against the published values [6]. In the third row $a\times b$ means $a$ parts of $b$ directions each, so that $\sum ab = K(k)$; a part of 3 is a zero-sum triple and a part of 2 an antipodal pair. The columns that disagree are treated in §F.2 and §F.3.

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| dimension | 25 | 26 | 27 | 28 | 29 | 30 | 31 |
| parts of $Z$ | 1×2 | 2×3 | 4×3 | 8×3 | 12×3+2×2 | 24×3 | 42×3 |
| (4) | 197 058 | 198 550 | 200 540 | 204 520 | 209 496 | 220 440 | 238 350 |
| published | 197 056 | 198 550 | 200 044 | 204 520 | 209 496 | 220 440 | 238 350 |
| | +2 | = | +496 | = | = | = | = |

The formula reproduces five independently obtained record configurations exactly. The columns $k = 1$ and $k = 3$ exceed the published value; §F.2 and §F.3 show that the additional points are available. The column $k = 2$ has no slack in this family; §F.4 leaves the family, and dimension 26 moves with it. Nor have the columns $k = 4$ to $k = 7$ any slack; §F.5 adds a fourth family at a different height, and dimensions 28 through 31 move with that. Every column of the table is therefore beaten, though four of them only by leaving the family the table describes.

C.4. **Dimensions 49–61 and 73–95.** Above dimension 48 the same construction runs over $P_{48}$ and $\Gamma_{72}$. Two things change. First, by Corollary C.2, $P_{48}$ admits only antipodal pairs while $\Gamma_{72}$ admits triples; this is the one structural difference between the two ranges. Second, the construction now needs many disjoint independent sets; the gain is proportional to their total size.

**Proposition C.3.** *In the one-point counts of Proposition B.8, the graph* $\Gamma_\gamma(\Lambda_{24})$ *has* 98 280 *vertices and is regular of degree* $n[2] = 4\,600$. *The graph* $\Gamma_\gamma(P_{48})$ *has* 26 208 000 *vertices and is regular of degree* $n[3] = 36\,848$. *The graph* $\Gamma_\gamma(\Gamma_{72})$ *has* 3 109 087 800 *vertices and is regular of degree* $n[3]+n[4] =$ 14 086 968. *By the Caro–Wei bound they contain independent sets of size at least* 22, 712 *and* 221.

Search on explicit coordinates finds independent sets of 248 lines in the first of these graphs, 7 069 in the second and 2 118 in the third. The ratio of the size found to the Caro–Wei bound is 11.3, 9.93 and 9.58.

Two ways of producing a disjoint family are used. The first takes automorphisms: if $g \in \mathrm{Aut}(L)$ and $C$ is independent then so is $Cg$, and two random images of a set of size $c$ overlap in about $c^2/\#\{\text{lines}\}$ lines, which we delete from the later image. This gives 1 400 pairwise disjoint independent sets of $P_{48}$, covering 8 023 414 lines or 30.6% of the minimal lines, and 32 000 of $\Gamma_{72}$, covering 47 150 230 lines. The second runs the search itself on a pool of minimal lines, removing each set from the pool as it is found; over $\Gamma_{72}$ it gives 32 000 sets covering 65 570 602 lines. The two have opposite shapes. The images of one good set begin at 2 106 lines and decay to 867, while the sets found afresh lie between 1 932 and 2 118. A dimension that reads few sets is better served by the first family and one that reads many by the second, the crossover falling at dimension 86. The two are never mixed, disjointness across families not being claimed.

**Theorem C.4.** *For $49 \le n \le 61$ and $73 \le n \le 95$ the lower bounds of Table 1 hold.*

*Proof sketch.* Apply (4) with $L = P_{48}$, $t = \frac{3}{4}$ and $|Z_i| = 2$ for $49 \le n \le 61$. For $73 \le n \le 95$ apply it with $L = \Gamma_{72}$, taking $t = \frac{2}{3}$ and $|Z_i| = 3$ where a partition of $Z$ into zero-sum triples is available, and $t = \frac{3}{4}$ with $|Z_i| = 2$ otherwise. The direction set $Z$ is a root system for $k \le 8$ and a cross-section of $\Lambda_{24}$ for larger $k$; if its partition has $r$ parts, take the $r$ largest available independent sets. At $t = \frac{3}{4}$ the poles are unconstrained and contribute $K(k)$; at $t = \frac{2}{3}$ we take for $W$ a rotated copy of $Z$ restricted to the points satisfying the 30° separation of (9), which is between 99% and 100% of it; it is all of $Z$ at $k = 3$, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15 and 16, and reaches $K(k)$ at $k = 3$, 5, 6, 7, 8, 9, 10, 14 and 16.

The choice of $Z$ is not forced. Conditions (7) and (8) ask of it only that it be a 60° code admitting a partition into parts internally at cosine $\le -\frac{1}{2}$, so a larger such code with a perfect partition is strictly better, and at $k = 12$ and 13 the cross-section taken was not the largest available. Let $\pi$ be an automorphism of the Golay code of cycle shape $1^6 3^6$, acting on $\Lambda_{24}$ by permuting coordinates. Its fixed space is 12-dimensional — six fixed points and six 3-cycles — and on the orthogonal complement, also 12-dimensional, $\pi$ has no eigenvalue 1, so $I + \pi + \pi^2 = 0$ there and every orbit $\{v, \pi v, \pi^2 v\}$ of a vector of that complement sums to zero. The 756 minimal vectors of $\Lambda_{24}$ lying in the complement have determinant $3^6$ at minimal norm 4, which is the Coxeter–Todd lattice $K_{12}$, and their partition into 252 zero-sum triples is the orbit set of $\pi$: a group acting, not a packing. Adjoining the one further direction carrying the most minimal vectors — 162 of them — gives a 13-dimensional section of 918, against $\Lambda_{12}$'s 648 and $\Lambda_{13}$'s 906. Thirteen is odd, so no order-3 isometry of $\mathbb{R}^{13}$ acts freely and those 306 triples are packed rather than given.

The same identity settles four partitions that a packing search had left short. An isometry $S$ with $I + S + S^2 = 0$ has eigenvalues $\omega$ and $\bar{\omega}$ only, so it acts freely and all its orbits are zero-sum triples; an $S$-invariant $Z$ is therefore partitioned perfectly by them, and those eigenvalues come in conjugate pairs, so this can happen only in even dimension. At $k = 16$, 18, 20 and 22 such an $S$ exists and the partitions improve from 1 434 triples and 7 pairs to 1 440 triples, from 2 457 and 11 to 2 466, from 5 782 and 23 to 5 800, and from 16 594 and 49 to 16 632. At $k = 20$ the isometry condition alone leaves the first level of the search 664 ways wide and a second invariant is needed: $S$ carries zero-sum triples to zero-sum triples, so it preserves the number of them through a vector, and filtering the candidate images by that degree narrows the level to eight. At $k = 22$ that degree is constant and the filter gives no further reduction; the isometry condition settles it there on its own, in twelve complete assignments. The six dimensions this moves are 84, 85, 88, 90, 92 and 94, by 549 638 spheres in total.

The odd $k$ carry no such $S$, and their partitions had been left short by a greedy that never revisits a placed triple: 9, 13, 25 and 60 triples at $k = 17$, 19, 21 and 23. What closes them is that the hypergraph of zero-sum triples is *linear* — two vectors lie in at most one triple, since the third is forced to be $-(a + b)$ — which determines the move set of a local search completely. If a triangle $\{u, v, w\}$ has $u$ uncovered and $v$, $w$ in a common part, that part's third vector is $-(v + w) = u$, which is uncovered; so it does not happen. Evicting one part therefore always leaves the number of uncovered vectors unchanged, and every neutral or improving move is a pair of uncovered vectors at 120°. Enumerating those pairs instead of sampling triangles takes the partitions from 1 773 triples and 11 pairs to 1 782, from 3 543 and 16 to 3 556, from 9 215 and 34 to 9 240, and from 30 990 and 82 to 31 050, perfect in every case, and moves dimensions 89, 91, 93 and 95 by 286 996 spheres.

The economy that suggests itself is not available. $Z = -Z$ and a triple's signs are fixed up to a global one, so its triples come in antipodal pairs and the search appears to halve — and an $S$-invariant partition is antipodally symmetric for the same reason, the orbit of $-v$ being the negative of the orbit of $v$. Summing $[a] + [b] + [c] = 0$ in $\Lambda/2\Lambda$ over such a partition gives $\sum[v] = 0$ over the $|Z|/2$ minimal lines, and for $\Lambda_k$ that fails at $k = 11$, 13, 15, 17, 19 and 23. On the vectors the same sum is doubled and says nothing, which is why the search runs there; a search on the lines stalls one triangle short at $k = 17$, at 890 of 891, for arithmetic reasons rather than for want of looking.

A copy of $Z$ cannot exceed $|Z|$, and at $k = 19$, 20 and 21 the direction set is the minimal-vector set of $\Lambda_k$, which is smaller than $K(k)$: 10 668, 17 400 and 27 720 against 11 948, 19 448 and 29 768. Those three layers were short by choice of ansatz rather than by geometry, since (9) and (10) ask of $W$ only that it be a 60° code lying 30° from $Z$ — not that it be a copy of anything. Taking for $W$ a rotated copy of the record configuration

of $\mathbb{R}^k$ instead gives 11 934, 19 415 and 29 712 poles. At $k = 19$ and 20 that configuration is available in the coordinates the direction set is already written in, $\Lambda_k$ there being $4\binom{n}{2}$ vectors $(\pm2, \pm2, 0, \dots, 0)$ together with the 128 even-sign patterns on each octad of a shortened Golay code; taking the octad signs odd and adjoining $((-1)^{c_1}, \dots, (-1)^{c_n})$ for $c$ in the dual of the octad code is Cohn–Li's construction, and it attains $K(k)$.

At $k = 21$ the direction set is written in 24 coordinates spanning 21, so a configuration built in $\mathbb{R}^{21}$ must be carried into the axis space by a rational similarity: $k$ orthogonal vectors of a common norm $\mu$ in the span. Whether one exists is decided by the discriminant of that space, since an orthogonal basis of norms $\mu$ has determinant $\mu^k$ modulo squares. At $k = 18$ and $k = 22$ the discriminant is 3 and $k$ is even, so none exists at all and no configuration built in the standard $\mathbb{R}^k$ can be placed there rationally; at $k = 21$ it is 1, so $\mu$ must be a square, which is why a frame drawn from the norm-32 direction vectors is always one short and one drawn from norm-4 vectors of the span is not.

That is a statement about the standard metric and not about the configuration. Three further direction sets are cross-sections written in more coordinates than they span, at $k = 15$, 17 and 18, and there the record configuration is not rebuilt but read: Cohn's coordinate data set gives it as exact integers together with an integer metric $\operatorname{diag}(c)$, which for $k = 15$, 17, 18 is $\operatorname{diag}(1^{14}, 2)$, $\operatorname{diag}(1^{16}, 2)$ and $\operatorname{diag}(1^{16}, 2, 6)$. A frame then has to satisfy $M^{\mathsf{T}}M = \mu \operatorname{diag}(c)$ and the discriminant condition becomes $\mu^k \det \operatorname{diag}(c) \equiv \operatorname{disc}$. At $k = 18$ that is $12 \equiv 3$, the discriminant of the axis space, and a frame does exist: sixteen columns $v_1 \pm v_2, \dots, v_{15} \pm v_{16}$ of norm 64 built from mutually orthogonal direction vectors, $2v_{17}$ of norm 128, and for the last one the orthogonal complement of those seventeen inside the span, which is a line and whose norm class is therefore forced — it comes out at 6 modulo squares, which is what the frame needs. The rotation is taken to be a product of rational plane rotations of $\operatorname{diag}(c)$, each of denominator $D$ with $a^2 + (c_j/c_i)b^2 = D^2$, since a full-rank Cayley transform on a space of this rank has denominator about $d^k$ and cannot be rounded finely enough to spend the margin. The layers go from 2 340, 5 338 and 7 374 to 2 562, 5 704 and 7 624 poles, against the published records 2 564, 5 730 and 7 654. At $k = 18$ that budget is not the best this paper knows: §E gives 8 358 points in dimension 18, some 700 more than the poles placed here. They are not used, and whether they can be is open. The configurations carried in above are read from [7] as integers against an integer metric, whereas the one of §E has its $\mathbb{R}^2$ part at multiples of 30° and so is defined over $\mathbb{Q}(\sqrt{3})$, not over $\mathbb{Q}$ in an 18-dimensional rational space. Dimension 83 below places a configuration realised over $\mathbb{Z}[\sqrt{2}]$, but there the directions and the poles lie in one ring; here they would not.

Dimension 83 is no longer an exception either. Its configuration is realised over $\mathbb{Z}[\sqrt{2}]$, but only the configuration is: a rational $Q = N/D$ with $Q^{\mathsf{T}}CQ = C$ carries the directions to poles in the same ring, and (9) becomes $(p +$

$q\sqrt{2})^2 \le 972D^2$ for integers $p, q$, decided exactly. All $604 = K(11)$ of them clear. □

Every dimension from 81 to 95 now carries its own construction: dimension 81 exceeds dimension 80 by 175 276, so no entry of Table 1 in this range rests on monotonicity of $K$. The construction also runs in dimensions 62 and 63, where it reaches 64 217 822 and 67 365 752 and is beaten; §D.1 treats those two.

The layered construction stops winning at 96. Throughout $n \le 95$ the Edel–Rains–Sloane total of §D stays below $\Gamma_{72}$. Its largest term is $A(n_0, \lceil n_0/4 \rceil)$, for which the code tables give at most $2^{30}$ up to $n_0 = 92$, or 17% of 6 218 175 600. At $n_0 = 93, 94, 95$ they give $2^{31}$, $2^{31}$ and $2^{32}$, and the whole chain evaluated at those three lengths gives $2.68 \times 10^9$, $2.68 \times 10^9$ and $4.83 \times 10^9$. At $n_0 = 96$ the code $[96, 33, 24]$ makes the leading term $A(96, 24) \ge 2^{33}$, at which the chain $(96, 24, 6, 1)$ totals 12 886 999 232, against the 6 480 558 568 the layered construction reaches there. Dimension 96 is therefore Edel–Rains–Sloane's rather than $\Gamma_{72}$'s, and §D.2 takes it on that footing — as §D.1 does dimensions 62 and 63.

Every input to those evaluations is a construction, so each is a lower bound for the Edel–Rains–Sloane count. A lower bound above $\Gamma_{72}$ settles which of the two constructions wins there; a lower bound below it settles nothing by itself. §6.1 measures the margin in the dimensions above 48.

*And the table stops there.* $K$ is superadditive: two $60^\circ$ codes placed on orthogonal subspaces meet at inner product 0, so $K(96 + m) \ge K(96) + K(m)$, and every dimension above 96 already follows from the row for 96 together with a published low-dimensional record. A row for dimension 100 reading $12\,886\,999\,232 + 24$ would be arithmetic on the row above it. What that leaves open is whether anything overtakes the implied floor above 96, as Edel–Rains–Sloane overtakes $\Gamma_{72}$ *at* 96. The layered construction does not: it ends at $k = 24$. The chain does, but by a margin that decides nothing — evaluated through $n = 128$ from the same tables used here, it exceeds the implied floor by at most 0.016%, at $n = 114$. So 96 is where the table stops saying anything, which is why it stops there.

## Appendix D. The Edel–Rains–Sloane chain

Four improvements are code-theoretic. The Edel–Rains–Sloane construction [19] builds a $60^\circ$ code from a chain $n \ge n_0 \ge 4n_1 \ge 16n_2 \ge \cdots$ of support sizes, giving

$$K(n) \;\ge\; \sum_{\nu} A(n, n_\nu, n_\nu)\, A\big(n_\nu, \lceil n_\nu/4 \rceil\big).$$

The construction is defined for every $n$, but its authors evaluate it only at $n = 32, 36, 40, 44, 64, 80, 128$. Elsewhere its value is a question about the current code tables rather than about the construction.

D.1. **Dimensions 62 and 63.** The first term of the chain needs no chain. A binary code of length $n_0$, dimension $m$ and minimum distance $d$ sends each codeword $c$ to the unit vector

$$x(c) \;=\; n_0^{-1/2}\big((-1)^{c_1},\dots,(-1)^{c_{n_0}}\big),$$

and two codewords at Hamming distance $j$ satisfy $\langle x(c),x(c')\rangle = (n_0-2j)/n_0$. Every pair is therefore at $60^\circ$ or more exactly when $d \ge \lceil n_0/4\rceil$; the $2^m$ sign vectors are then a kissing configuration of $\mathbb{R}^{n_0}$, hence of $\mathbb{R}^n$ for every $n \ge n_0$:

$$K(n) \;\ge\; \max_{n_0\le n} A\big(n_0,\lceil n_0/4\rceil\big).$$

Since $\lceil 62/4\rceil = \lceil 63/4\rceil = 16$, the codes $[62,26,16]$ and $[63,27,16]$ of [22] qualify, giving $2^{26}$ and $2^{27}$ sign vectors. Both are shortenings of one $[64,28,16]$; shortening an $[n,m,d]$ code gives $[n-1,m-1,\ge d]$. That alone already beats the values of §1.2.

The rest of the chain is not negligible: it carries Edel–Rains–Sloane's own dimension-64 total from $2^{28} = 268\,435\,456$ to $331\,737\,984$. Since $4\cdot 16 = 64 > 63$, the largest level-one weight the chain allows in either dimension is $n_1 = 15$, and $15\cdot 4 = 60 \le 62$, so the chain is $(n,15,2)$ and

$$K(n) \;\ge\; A\big(n,\lceil n/4\rceil\big) + A(15,4)\,A(n,15,15) + 4\tbinom{n}{2}.$$

**Theorem D.1.** *$K(62) \ge 71\,310\,732$ and $K(63) \ge 138\,419\,844$.*

*Proof.* Level zero gives $2^{26}$ and $2^{27}$ as above. For level one, index the cells of a $15\times 4$ grid by 60 of the coordinates and let a support take one cell in each row; two supports then meet in exactly $15-d_H$ of their choice words, so a quaternary code of length 15 and minimum distance 8 yields supports meeting in at most $7 = \lfloor 15/2\rfloor$. The cyclic $[15,6,8]_4$ with defining set $\{0,1,2,3,4,5,8,10,12\}$, a union of cyclotomic cosets modulo 15 under $x\mapsto 4x$, gives $A(n,15,15) \ge 4^6 = 4\,096$, and $A(15,4) = 1\,024$ from the extended Hamming $[16,11,4]$ shortened once. Level two takes every pair of coordinates with all four sign patterns. Hence

$$67\,108\,864 + 4\,194\,304 + 7\,564 = 71\,310\,732,$$
$$134\,217\,728 + 4\,194\,304 + 7\,812 = 138\,419\,844. \qquad \square$$

Everything above level zero is built rather than cited: [26] enumerates all $4^6$ codewords of the quaternary code to establish its minimum distance, checks all $\binom{4096}{2}$ support overlaps directly, enumerates all $2^{10}$ codewords of the $[15,10,4]$, and runs the whole chain end to end in coordinates at the two dimensions where it happens to reproduce the exact kissing number — $n = 8$, chain $(8,2)$, giving the 240 points of the $E_8$ root system, and $n = 16$, chain $(16,4,1)$, giving 4 320, the Barnes–Wall value and the record — with every pair of both configurations checked. Only the two level-zero code parameters are taken from [22].

Against the values of §1.2 these are factors 1.36 and 2.64. Level zero alone already exceeds the *ceiling* of the construction of §C.4 in these two

dimensions, so no family of independent sets can recover them there. Over $P_{48}$ that construction gives $52\,416\,000 + 2\sum_i |C_i| + K(k)$ by (4) with $|Z_i| = 2$. The number of independent sets is at most the number $\lfloor K(k)/2 \rfloor$ of antipodal pairs available among the directions, and each $|C_i|$ is at most the largest independent set known, of 7 069 lines. At $k = 14$ and $k = 15$ that gives 66 075 240 and 70 543 480, short of $2^{26}$ and $2^{27}$ by 1 033 624 and 63 674 248. Closing those gaps would need independent sets of 7 605 and 31 903 lines against the largest known 7 069, or more parts than the record in dimension 14 or 15 allows. The chain is not a refinement of §C.4 in these dimensions but a different and better construction.

D.2. **Dimension 96.** At $n = 96$ the chain closes up: $96 \geq 96 \geq 4 \cdot 24 \geq 16 \cdot 6 \geq 64 \cdot 1$, so $(96, 24, 6, 1)$ is admissible, and every inequality but the last is an equality. Level zero is again a single sign code, and here it carries two thirds of the total: $\lceil 96/4 \rceil = 24$, and the code $[96, 33, 24]$ of [22] gives $A(96, 24) \geq 2^{33}$.

The two middle levels apply the grid map of §D.1 at two different factorisations of 96. Writing $96 = 24 \cdot 4$ and indexing the cells of a $24 \times 4$ grid by the coordinates, a support taking one cell in each row has weight 24, and two supports meet in exactly $24 - d_H$ of their choice words; a quaternary code of length 24 and minimum distance 12 therefore gives supports meeting in at most $12 = \lfloor 24/2 \rfloor$, and $[24, 9, 12]_4$ of [22] gives $A(96, 24, 24) \geq 4^9$. Writing instead $96 = 6 \cdot 16$, the same argument over $\mathbb{F}_{16}$ with the Reed–Solomon code $[6, 4, 3]_{16}$ gives $A(96, 6, 6) \geq 16^4$.

**Theorem D.2.** $K(96) \geq 12\,886\,999\,232$.

*Proof.* The four levels contribute

$$A(96, 24) + A(96, 24, 24)\, A(24, 6) + A(96, 6, 6)\, A(6, 2) + A(96, 1, 1)\, A(1, 1),$$

in which $A(6, 2) = 32$, $A(1, 1) = 2$ and $A(96, 1, 1) = 96$ are exact, while [4] brackets $A(24, 6)$ between 16 384 and 24 106; a larger value would only raise the total, so the smaller end is substituted. With that and the three lower bounds above,

$$8\,589\,934\,592 + 262\,144 \cdot 16\,384 + 65\,536 \cdot 32 + 96 \cdot 2 \;=\; 12\,886\,999\,232. \qquad \square$$

The value of §1.2 here is the direct sum $\Gamma_{72} \oplus \Lambda_{24}$, at $6\,218\,175\,600 + 196\,560$, so the factor is 2.07. As in §D.1 the mathematics is Edel–Rains–Sloane's, and as there we find no record of their construction having been evaluated at this length, which is not among their seven worked examples; their own first remark licenses substituting any available lower bounds for the code sizes, which is all that is done above. The total is a function of those tables: with the three levels below weight 96 as constructed here — the weight-24, weight-6 and weight-1 terms, together 4 297 064 640 — the chain passes the direct sum as soon as $A(96, 24) \geq 2^{31}$, and $[96, 33, 24]$ gives $2^{33}$. Of the three code inputs only the smallest is built here: [26] constructs $[6, 4, 3]_{16}$ and establishes its minimum distance by evaluating all $16^4$ polynomials.

The sign code $[96, 33, 24]$, which carries two thirds of the total, and the quaternary code $[24, 9, 12]_4$ are taken from [22].

## Appendix E. Dimension 18: a hexagon of bent cosets

The one improvement below dimension 25 uses none of the three extremal lattices above, and no code chain either. Write a point of $\mathbb{R}^{18} = \mathbb{R}^{16} \oplus \mathbb{R}^2$ as $(v, p)$ in units where the norm is 8, so that two points are compatible exactly when $\langle (v,p), (v',p') \rangle \le 4$. The configuration is a hexagon:

Table 4. The dimension-18 configuration. The first three rows are the layout of [12]; the whole improvement is in the fourth.

| | $\lvert p\rvert^2$ | angles of $p$ | $v$ | points |
|---|---|---|---|---|
| equator | 0 | — | odd $\mathrm{BW}_{16}$ | $480 + 30 \cdot 128 = 4\,320$ |
| tiers | 2 | $0°, 60°, \dots, 300°$ | $\pm 1$ on a 6-set of that angle's family | $6 \cdot 16 \cdot 32 = 3\,072$ |
| poles | 8 | $0°, 60°, \dots, 300°$ | 0 | 6 |
| tier B | 8/9 | $30°, 90°, \dots, 330°$ | $(2/3)(-1)^c$, $c$ a sign word | $6 \cdot 160 = 960$ |
| total | | | | 8 358 |

Here odd $\mathrm{BW}_{16}$ means the 480 vectors $(\pm 2, \pm 2, 0^{14})$ together with the vectors supported on an octad of $\mathrm{RM}(1,4)$ with entries $\pm 1$ and an odd number of minus signs. Decoding the 7 654 of [12] exhibits exactly this hexagon, with their outer tier occupying the two angles 0° and 180° and carrying 256 vectors. The gain of +704 is entirely in that outer tier, and it comes from choosing the three six-set families differently.

E.1. **What the outer tier costs.** Let $c \in \mathbb{F}_2^{16}$ be the sign word of a tier-B vector. Against the odd octad vectors $c$ must be even on every octad, that is $c \in \mathrm{RM}(2,4)$; against a tier at angular distance less than 90° it must be even on every six-set of that tier's family; at 90° or more there is no constraint at all, which is why the six angles $30° + 60°k$ are better than the two that [12] use. For $c \in \mathrm{RM}(2,4)$ the parity $\langle c, \mathbf{1}_X \rangle$ depends only on the coset of $\mathrm{RM}(1,4)$ containing $\mathbf{1}_X$. A family whose sixteen six-sets all lie in *one* coset $z + \mathrm{RM}(1,4)$ therefore costs tier B a single linear condition $\langle c, z \rangle = 0$, whereas a family spread over two cosets costs two.

The cosets of $\mathrm{RM}(1,4)$ inside $\mathrm{RM}(2,4)$ that contain weight-6 words are exactly the 28 *bent* cosets, of weights 6 and 10, sixteen words each. The sixteen weight-6 words of a bent coset form a symmetric 2-$(16, 6, 2)$ design, so they pairwise meet in two cells; that is a valid tier, with sixteen members.

E.2. **Which hexagons are open.** On $\mathrm{RM}(2,4)/\mathrm{RM}(1,4) \cong \mathbb{F}_2^6$ the pairing $B(q,z) = \langle q, z \rangle$ is a nondegenerate symplectic form, and "bent" is the quadric $O^+(6,2)$ of 28 nonsingular points. For bent $q$ and $z$ one has

$B(q, z) = 1$ if and only if $q + z$ is bent — and that is also exactly the condition for two tiers $60°$ apart to be compatible, their six-sets meeting in at most three cells. A hexagon of tiers is therefore a triangle $z_0, z_1, z_2$ in the bent-sum graph, and there are two kinds.

If $z_2 = z_0 + z_1$, the triangle is *closed*: all twelve possible tier-B angles see the same 512-word space, and since a word may be used at only one angle the outer tier is capped at 512.

If $z_0 + z_1 + z_2 \notin \mathrm{RM}(1, 4)$ — *open*, which 280 of the 336 triangles are — the six angles $30° + 60°k$ see three *different* 512-word spaces, pairwise meeting in a 256-word one. Each is an affine 4-space of $\mathrm{RM}(1, 4)$-cosets over an elliptic quadric, and full cosets pairwise at bent difference form a distance-6 code: the bent-difference graph on those sixteen cosets is the complement of the Clebsch graph, whose clique number is 5, so each angle carries $5 \cdot 32 = 160$ words. The six sets of 160 are pairwise disjoint, since distinct words of $\mathrm{RM}(2, 4)$ are automatically at distance at least 4, which is all that angles $60°$ or more apart require. This gives $6 \cdot 160 = 960$ and

$$K(18) \;\geq\; 4\,320 + 3\,072 + 6 + 960 \;=\; 8\,358.$$

The first family of [12] is one of these 28 bent cosets; their other two are each half of one bent coset and half of another. That costs an extra linear condition apiece and makes every angle see the same 256 words, so 256 was indeed a ceiling — for their base, with the code varying, rather than for the construction.

## Appendix F. Historical constructions and the frozen September table

The following material records the eight earlier constructions that are no longer state of the art in the reviewed sources. It is retained for reproducibility and attribution; the main text reports the current table only. In that historical split, forty-three of the fifty-three entries lay above dimension 48.

F.1. **Superseded construction descriptions.** The following detailed accounts are retained as historical documentation.

F.2. **Dimension 25: the choice of level.**

**Theorem F.1.** $K(25) \geq 197\,058$.

*Proof.* Take $L = \Lambda_{24}$, $k = 1$, one independent set $C$ of 248 lines, and $Z = W = \{+1, -1\} \subset S^0$. The published configurations use $t = \frac{2}{3}$, at which $b = 1/\sqrt{3} = 0.577\ldots > \frac{1}{2}$; then (9) fails for the two poles, which are discarded.

The level $t = \frac{2}{3}$ is not forced here. Constraint (11) bounds it from below and (7) from above: for two antipodal directions the latter requires $-1 \leq (\frac{1}{2} - t)/(1 - t)$, that is $t \leq \frac{3}{4}$. At $k = 1$ the sphere $S^0$ has two points and admits no triple; the whole range $\frac{2}{3} \leq t \leq \frac{3}{4}$ is therefore admissible.

At $t = \frac{3}{4}$ constraint (6) becomes $\frac{3}{4}\gamma + \frac{1}{4} \le \frac{1}{2}$, i.e. $\gamma \le \frac{1}{3}$; and for $\Lambda_{24}$ the thresholds $\gamma = \frac{1}{4}$ and $\gamma = \frac{1}{3}$ define the same condition, since Lemma 3.4 allows only $\langle u, u' \rangle / \mu \in \{0, \pm\frac{1}{4}, \pm\frac{1}{2}\}$ and no admissible value lies strictly between $\frac{1}{4}$ and $\frac{1}{3}$. The independent set is therefore unchanged at 248 lines. At $t = \frac{3}{4}$ we have $b = \frac{1}{2}$, so (9) holds and both poles are admissible. Finally (4) gives $196\,560 + 2 \cdot 248 \cdot (2 - 1) + 2 = 197\,058$. □

Three constraints meet at $t = \frac{3}{4}$. The bound $t \ge \frac{2}{3}$ keeps the independent set at 248 lines, the bound $t \le \frac{3}{4}$ keeps the antipodal pair admissible, and the poles require $t \ge \frac{3}{4}$. The argument does not repeat for $k \ge 2$: triples exist there and contribute twice as much as pairs, which makes $t = \frac{2}{3}$ optimal; the poles already attain $K(k)$ at that level.

The gain is $+2$ over any independent set used in its place. The level argument uses the 248-line independent set of [34] unchanged; one of 249 lines would give 197 060 by the same argument.

*Superseded, 2026-09-07.* The current dimension-25 entry no longer rests on this level argument. In joint work in progress with H. Cohn and B. Lindow the caps are rebuilt from *heads*: points $x$ of squared length 3 (norm-4 units) lying in the lens of a minimal vector, each removing exactly its owner from the equator. Two compatible heads never share a removal, so the count is exact: the equator loses its $H$ owners and gains the $2H$ points $\pm x$, the two poles are admissible, and one further equator point $-\frac{2}{\sqrt{6}}v$ that is not a lattice vector is added, giving $196\,560 - H + 1 + 2H + 2 = 196\,560 + H + 3$.

The heads come in blocks. Fix a vector $v \in \Lambda_{24}$ of norm 6, take the 552 minimal vectors $z$ with $\langle z, v \rangle = -3$, and push each of them towards $v$ by $t_- = (3 - \sqrt{3})/6$. Every pair of heads within one block then meets the bound $\langle x, x' \rangle = 1$ exactly, so a block is a tight packing. Its size is optimal: a block projects to a code of pairwise cosine $\frac{1}{5}$ in $\mathbb{R}^{23}$, and the largest such code has 552 points. Searched over the whole norm-6 shell of 16 773 120 vectors, a union of blocks reaches 971 heads — three vectors $v$ with pairwise inner products $-4$, 0 and $-3$ — and a fourth vector contributes exactly one head more. Every remaining minimal vector that can own a head over that core contributes one candidate, the point of largest norm in its own polytope. That pool is complete at 1 188 heads over 1 188 distinct owners. The supplement is an independent set in the conflict graph the pool carries, found by maximising over that graph rather than by growing the set head by head. The shipped configuration takes 45 of them, so $H = 1\,016$ and $K(25) \ge 197\,579$, verified in exact arithmetic by the package `dim25-lens-heads`. The level argument above remains the template's own bound and is kept as such.

### F.3. **Dimension 27: partitioning the directions.**

**Theorem F.2.** *$K(27) \ge 200\,540$.*

*Proof.* Take $L = \Lambda_{24}$, $k = 3$ and $t = \frac{2}{3}$. Take $Z$ to be the $A_3$ root system normalised to the sphere, the 12 vertices of a cuboctahedron and a kissing configuration of $\mathbb{R}^3$, and take $W$ to be a rotated copy of it satisfying the $30^\circ$ separation of (9), so that $|W| = K(3) = 12$. By Corollary C.2 each independent set may carry a zero-sum triple. The 12 vertices admit a partition into four zero-sum triples; four independent sets therefore suffice, and (4) gives

$$196\,560 + 2 \cdot 4 \cdot 248 \cdot (3-1) + 12 \;=\; 196\,560 + 3\,968 + 12 \;=\; 200\,540. \quad \square$$

The choice of direction set is constrained, and not merely convenient. The 12-point kissing configuration of $\mathbb{R}^3$ is not unique; two of them are classical, the cuboctahedron and the icosahedron. Icosahedral vertices meet at $\langle z, z' \rangle \in \{\pm 1, \pm 1/\sqrt{5}\}$, so no two of them that are not antipodal reach the $-\frac{1}{2}$ that Corollary C.2 demands: the icosahedron admits no zero-sum triple at all, and over it the construction would be reduced to six antipodal pairs and 199 548 points. The cuboctahedron has 24 pairs at $-\frac{1}{2}$ and admits exactly two partitions into four triples.

The published configuration [34] groups its twelve directions into parts of sizes $3, 3, 2, 2, 2$. That partition uses five independent sets and deletes $5 \times 496 = 2\,480$ vectors from the equator; four triples use four and delete 1 984, returning 496 vectors. The construction adds no new spherical code: the Leech vectors, the twelve directions, the twelve poles and four of the five independent sets are those of the published configuration; only the assignment of independent sets to directions differs. A script in [26] regenerates the configuration from the dimension-27 block of [7]. B. Lindow found the same construction independently (private communication through H. Cohn, September 2026).

The family admits nothing further in this dimension. By (4) the gain is $2\sum_i |C_i|(|Z_i| - 1)$ with $|Z_i| \le 3$ and $\sum_i |Z_i| \le |Z| = 12$, so $\sum_i (|Z_i| - 1)$ is at most $\lfloor 2 \cdot 12/3 \rfloor = 8$, which four triples attain.

*Superseded, 2026-09-08.* The current dimension-27 entry no longer rests on Theorem F.2; see §F.4. The theorem and the partition argument remain the template's own bound, and the twelve directions they fix are the ones the new configuration uses.

### F.4. **Dimensions 26 and 27: the coset triangle on every triangle of directions.**

**Theorem F.3.** *$K(26) \ge 199\,806$ and $K(27) \ge 201\,566$.*

Both are joint work in progress with H. Cohn and B. Lindow, and the account is the separate note of §F.2; we record the shape and the verification. Write a point of $\mathbb{R}^{24+k}$ as $(x, y)$ with $y \in \mathbb{R}^k$, normalise every point to squared length 4 so that compatibility is $\langle P, Q \rangle \le 2$, and put the $K(k)$ axis points $(0, a)$, $|a| = 2$, on a hexagon ($k = 2$) or a cuboctahedron ($k = 3$). A cap point $(x, y)$ then has $|y| \le \sec\theta$ for the covering radius $\theta$ of the axis set;

for the hexagon $\theta = 30°$ forces $|x|^2 \geq 8/3$ with equality forcing $y$ onto the six edge midpoints, a head at $|x|^2 = 8/3$ carries a zero-sum triangle of directions, and two cap points there are compatible iff $\langle x, x' \rangle \leq 2 - \frac{4}{3}\cos\angle(y, y')$: $\frac{2}{3}$ on a common direction, $\frac{4}{3}$ at 60°, no condition at 120° or 180°. So the layer is one head set per triangle of directions — two for the hexagon, four for the cuboctahedron, whose twelve vertices are four zero-sum triangles every two of which have a pair at 60° — each a code of cosine at most $\frac{1}{4}$ on the sphere $|x|^2 = 8/3$, cross-constrained at cosine $\frac{1}{2}$, and the count is $196\,560 + K(k) + 2(P - F) + 3F$ for $P$ heads of which $F$ remove nothing.

Every head is $y/3$ with $y = 3u+v$ integral, $u$ a minimal vector (the owner, which the head removes) and $v$ a Leech vector of norm 6 with $\langle u, v \rangle = -3$, or a free head $y = \pm 2v$; a head removes exactly its owner, and two heads on one owner have $\langle y, y' \rangle \geq 16$, above every threshold, so removals are not shared. Three such $v$ with $v_1 + v_2 + v_3 = 0$ carry 1656 heads over three disjoint blocks $\Sigma(v_i) = \{u : \langle u, v_i \rangle = -3\}$ of 552, and on one side the exact optimum is 762 of them, against 552 for a single $v$, which is Cohn's construction. In dimension 26 the same triangle serves both sides: with $\iota(u) = -v_i - u$, a fixed-point-free involution of each block, $\langle 3u + v_i, 3\iota(u') + v_i \rangle = 15 - 9\langle u, u' \rangle$, which respects the cross threshold 12 exactly when $\langle u, u' \rangle \geq 1$; so a side $S$ and its image $\iota(S)$ fit on the two triangles of the hexagon whenever the owners of $S$ are pairwise at inner product 1 or 2 inside each block, and an optimal $S$ with that property exists. The shipped configuration, found by a solve in which each head chooses its side, has 1458 class heads, 486 over each lean of the triangle $v_1, v_2, v_3$; its 96 free heads are given up for the second cap layer described below, which they alone were blocking, and $196\,560+6+2\cdot1458+324 = 199\,806$. A free head contributes 3 compared with 2 for a class head and removes nothing, so admitting the free layer requires deleting class heads that conflict with it. The conflicts concentrate: about 72 class heads block the whole norm-6 shell, and 489 candidates share one of them. The two choices therefore have to be made together, by maximizing $3|F| - \sum_h \mathrm{cost}(h)$ over the deleted heads $h$. Evaluating one candidate at a time makes every free head appear unaffordable. In dimension 27 the axis is the cuboctahedron rotated by 45° about a coordinate axis, whose largest cosine to a direction is $(2+\sqrt{2})/4$ and whose inner product with a cap point is at most $(2+\sqrt{2})/\sqrt{3} < 2$; two triangles of leans with cross Gram matrix the circulant of $(3, 0, -3)$ carry the class heads over the four sides, alongside free heads that remove nothing. The three kinds of point contribute differently: a class head contributes +2, a free head +3, and a second-layer head +1; they compete for the same owners, so they have to be chosen *together*. Choosing them in sequence, as the first version of this construction did, gives 2133 class heads, 129 free heads and 284 second-layer heads. Choosing all three at once, which is B. Lindow's contribution, gives 2090, 168 and 310: 43 class heads are removed, at −86, to admit 39 free heads at +117 and 26 second-layer heads at +26, for a net +57. The sequential search

had already discarded this joint modification because each isolated change decreases the count. Both dimensions carry a second cap layer, at $|x|^2 = 3$ with $|y| = 1$; what separates them is the number of admissible points. Such a cap satisfies $\langle y, a\rangle \le 2$ whatever direction it points, since $|y||a| = 2$; it carries the antipodal pair $\pm y$ and no more, three directions needing pairwise cosine $\le -1$; and the scaled owner $x = \frac{\sqrt{3}}{2}u$ deletes exactly $u$, because $\langle x, z\rangle > 2$ forces $\langle u, z\rangle > 4/\sqrt{3}$, that is $\langle u, z\rangle = 4$. Two cap points for one deletion: $+1$ each. Its direction must clear the first-layer directions, whose covering radii differ — 30° for the hexagon, 45° for the cuboctahedron — so a first-layer head may meet a second-layer owner at $\langle Y, u\rangle \le 24$ in dimension 26 and at $\langle Y, u\rangle \le 32$ in dimension 27. A free head $y = \pm 2v$ has $\langle Y, u\rangle = 2\langle v, u\rangle$, a multiple of 16: it clears the bar 32 and can never clear the bar 24. Dimension 27 therefore keeps its free heads and its second layer at once, 310 heads over 3 110 admissible owners, while dimension 26 can open the layer only by removing every free head. The count above loses $96 \cdot 3$ points and adds 324 second-layer points, for a net $+36$. That layer is optimal and not merely the best found: its 1 188 admissible owners carry a graph under $\langle u, u'\rangle = 16$ that is exactly 64-regular, constraint programming closes its independence number at 108, and the three lines take three disjoint such sets, so 324 is the maximum. Surrendering dimension 27's free heads in the same way enlarges its admissible set from 3 009 to 4 634 and does not recover the loss of 387 points. Its direction must clear the twelve first-layer directions, and that is where the two dimensions part: the hexagon's six edge midpoints leave a gap of 30°, at which not one of the 196 560 minimal vectors qualifies, while the cuboctahedron's covering radius is 45°, attained at its six square-face centres $\pm e_1, \pm e_2, \pm e_3$ – three antipodal lines. One quantum, and the admissible owners go from 0 to 3110. Over that first layer the second layer is again an independent-set problem, and 310 of them fit, $105 + 104 + 101$, jointly with the first layer rather than over a first layer already fixed — which is what the joint optimisation admits, since on a first layer optimised alone only 105 of them still fit. Everything reduces to $\langle Y, u\rangle \le 32$ and $\langle u, u'\rangle \le 8$ on a line, since $\frac{\sqrt{3}}{48} \cdot 32 + \frac{2}{\sqrt{6}} = (2 + \sqrt{2})/\sqrt{3} < 2$ and $\frac{3}{32} \cdot 8 + 1 = \frac{7}{4} < 2$: $196\,560 + 12 + 2 \cdot 2090 + 3 \cdot 168 + 310 = 201\,566$. The package `dim26-27-iota-triangles` stores the heads as integer vectors with the index of their triangle, rebuilds the Leech vectors from the Golay code, decides every head–equator and head–head pair by integer comparison against the thresholds $144 - 96c$ ($c$ the exact largest cosine between two triangles of directions, in the scaling with minimum 32) and the axis inequalities in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and rejects a head moved, duplicated or displaced.

### F.5. **Dimensions 28 to 31: a second layer, at a free height.**

**Theorem F.4.** *$K(28) \ge 204\,896$, $K(29) \ge 209\,968$, $K(30) \ge 221\,012$ and $K(31) \ge 238\,662$.*

These are joint work in progress with H. Cohn and B. Lindow, and the account is the separate note of §F.2; as in §F.4 we record the shape and the verification. Table 3 is exact in the columns $k = 4$ through $k = 7$, so the template of §C.3 has no slack in any of them: every point beyond the published value has to come from outside (3).

Normalise every point of $\mathbb{R}^{24+k}$ to squared length 4, so that compatibility reads $\langle P, Q\rangle \le 2$, and write it as $(x, y)$ with $y \in \mathbb{R}^k$. In these units the three families of (3) are the equator $(u, 0)$, the caps $(\sqrt{2/3}\,u, (2/\sqrt{3})z)$ and the axis $(0, 2a)$, where $u$ runs over the minimal vectors of $\Lambda_{24}$, now of squared length 4.

Proposition C.1 holds every lifted point at one level, and it is (6) — two caps on distinct lines of one $C_i$ at a common direction — that forces $t \ge 1/(2(1-\gamma)) = \frac{2}{3}$. That constraint exists because a whole independent set rides each direction. A family that places only a *few* points above each direction does not meet it, and for such a family the level is free. Write its points as $(x, hw)$ with $|w| = 1$ and $|x|^2 + h^2 = 4$, and call $h$ the *height* and $x$ the *head*. Four thresholds move with $h$:

$$\begin{array}{ll} \text{no equator point is deleted} & \langle x, v\rangle \le 2 \text{ for every minimal } v, \\ \text{an axis point survives} & \langle w, a\rangle \le 1/h, \\ \text{two directions of one head} & \langle w, w'\rangle \le 1 - 2/h^2, \\ \text{two heads on one direction} & \langle x, x'\rangle \le 2 - h^2, \end{array} \tag{12}$$

and against a cap the condition is

$$\sqrt{2/3}\,\langle x, u\rangle \;+\; \frac{2h}{\sqrt{3}}\,\langle w, z\rangle \;\le\; 2, \tag{13}$$

with $u$ an owner and $z$ a direction of the part carrying it. The last line of (12) holds the number of heads on one direction to $2/(2 - |x|^2)$ when $|x|^2 < 2$: the unit vectors $x/|x|$ are then pairwise at cosine at most $-c$ with $c = (2 - |x|^2)/|x|^2$, and unit vectors so placed number at most $1 + 1/c$. The first line is the key property of such a layer is that it it does not remove an equator point; its additions are therefore limited to axis points and classes.

Lattice facts supply the heads. If $g \in \Lambda_{24}$ has $|g|^2 = 2r$ and $v$ is minimal, then $g - v$ has squared length $2r + 4 - 2\langle g, v\rangle$, an even integer that is 0 or at least 4; it is 0 only for $g = v$, and the value $\langle g, v\rangle = r + 1$ would make it 2, which no lattice vector has. Hence $\langle g, v\rangle \le r$: at most 3 for a head of squared length 6 and at most 4 for one of squared length 8. If $|x|^2 = 1$, so that $x = u_0/2$ for a minimal $u_0$, Cauchy–Schwarz gives $\langle x, v\rangle \le 2$ with equality only at $v = u_0$. So $x = v/2$ with $|v|^2 = 6$, $x = f/\sqrt{3}$ with $|f|^2 = 6$, $x = v/2$ with $|v|^2 = 8$ and $x = u_0/2$ all satisfy the first line of (12), at heights $h^2 = \frac{5}{2}$, 2, 2 and 3; these are the four that occur. The norm of the head is a second free parameter, the two entries at $h^2 = 2$ differ only in it, and it is the norm-8 one at $h = \sqrt{2}$ that carries all four dimensions below. One consequence of (13) is useful to record before the cases, because it does not depend on the height: if the direction equals the axis threshold,

$\langle w, z\rangle = 1/h$, the second term is $2/\sqrt{3}$ whatever $h$ is, and the condition collapses to

$$\langle x, u\rangle \;\le\; \sqrt{6} - \sqrt{2} \;=\; 1.0352\ldots \tag{14}$$

TABLE 5. The layer of Theorem F.4, in the normalisation $|P|^2 = 4$. The head $x$ is the $\mathbb{R}^{24}$ part of a layer point and $w$ its direction; $z$ is a cap direction. All four columns are one layer, the norm-8 frame at height $\sqrt{2}$: the head runs over all 48 vectors of a Leech frame and the direction over the whole cross-polytope of $\mathbb{R}^k$, which is $96k$ points. The axis is the maximum 60° code of [34] that Table 3 reproduces, carried into the frame's coordinates — and in dimensions 30 and 31 rotated first, which is free and leaves the layer unchanged. No layer point deletes an equator point, and the last three lines give the full count.

| $k$ | 4 | 5 | 6 | 7 |
|---|---|---|---|---|
| dimension | 28 | 29 | 30 | 31 |
| height $h$ | $\sqrt{2}$ | $\sqrt{2}$ | $\sqrt{2}$ | $\sqrt{2}$ |
| $\lvert x\rvert^2 = 4 - h^2$ | 2 | 2 | 2 | 2 |
| $\lvert\text{head}\rvert^2$ | 8 | 8 | 8 | 8 |
| head $x$ | $v/2$ | $v/2$ | $v/2$ | $v/2$ |
| directions $w$ | $\pm e_{1..4}$ | $\pm e_{1..5}$ | $\pm e_{1..6}$ | $\pm e_{1..7}$ |
| axis threshold $1/h$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ |
| largest $\langle w, z\rangle$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ | $1/\sqrt{2}$ |
| directions per head | 8 | 10 | 12 | 14 |
| heads per direction | 48 | 48 | 48 | 48 |
| layer points | 384 | 480 | 576 | 672 |
| axis points kept | 16 of 24 | 32 of 40 | 68 of 72 | 118 of 126 |
| owner lines | $8 \cdot 248$ | $14 \cdot 248$ | $24 \cdot 248$ | 10 328 |
| net gain | +376 | +472 | +572 | +312 |

*Dimensions 28, 29 and 30: one layer.* Take $h = \sqrt{2}$ and the head from a lattice vector of squared length 8: $x = v/2$, so that $\langle x, u\rangle \le 2$ with equality and the layer deletes nothing. Three of its four constraints are then free *in every dimension*. The third line of (12) reads $\langle w, w'\rangle \le 0$, so one head carries a whole cross-polytope, $2k$ directions. The fourth reads $\langle x, x'\rangle \le 0$, so two heads share a direction as soon as their lattice vectors are non-acute — and the 48 vectors $\pm f_1, \ldots, \pm f_{24}$ of a frame satisfy that throughout, while 48 vectors of $\mathbb{R}^{24}$ pairwise at non-positive inner product *are* a frame, so 24 head lines is the most there is. The layer is

$$48 \times 2k \;=\; 96k \quad \text{points}, \tag{15}$$

384, 480, 576, 672 in dimensions 28, 29, 30, 31.

What is left is (13), and it is the whole content. If every owner is *type B* for the frame — $|\langle u, f_i\rangle| = 2$ for all 24 heads, never 3 or 4 — the threshold on the directions is

$$\langle z, w\rangle \;\le\; \sqrt{3/2} - \tfrac{1}{2} \;=\; 0.7247448\ldots, \tag{16}$$

and the $\tau(k)$ cap directions are a root system: $D_4$, $D_5$, $E_6$, $E_7$ each have an orthonormal frame in which *every* root has largest coordinate $1/\sqrt{2} = 0.7071067\ldots$ For $E_7$ one has to rotate a pair of basis vectors by 45°, since two of its roots are otherwise aligned with a frame vector. So the margin is a single inequality, and it is the one the norm of the head permits:

$$1/\sqrt{2} \le \sqrt{3/2} - \tfrac{1}{2} \quad\Longleftrightarrow\quad 1 + 1/\sqrt{2} \le \sqrt{3} \quad\Longleftrightarrow\quad \sqrt{2} < \tfrac{3}{2}. \tag{17}$$

A norm-6 head at the same height would need $\langle f, u\rangle \le 1$ instead, which forbids 11 730 minimal lines to each head where the norm-8 head forbids the 2 094 at $|\langle v, u\rangle| \ge 3$.

So the layer costs nothing geometrically and everything in the classes: every one of them must be type $B$ for one common frame — 8 classes in dimension 28, 24 in dimension 30, 42 in dimension 31, and 14 in dimension 29, which carries twelve zero-sum triangles and two antipodal pairs because $\tau(5) = 40$ is not a multiple of 3. This follows from the following coordinate choice. Choose the coordinates so that the frame is $8e_1, \ldots, 8e_{24}$; then the type-$B$ lines are exactly the $759 \times 64 = 48\,576$ octad vectors $(\pm 2^8, 0^{16})$ of the Golay code, and the whole monomial group $2^{12}{:}M_{24}$ permutes them. *Every monomial image of a type-B class is again a type-B class*, so nothing has to be filtered and what is left is a packing. The remaining moves are explicit as well: a type-$B$ line is an octad together with a position in $\mathbb{F}_2^6$, a Golay sign word acts on it by translating that position inside its octad, and the sign word of a class is therefore an exactly solvable coordinate. Coordinate descent over the sign words, with a fresh batch of permutations at each visit to a class, gives 14 pairwise disjoint 248-line classes in dimension 29 — the full $14 \cdot 248$, with nothing given up. Dimension 30 is exact as well, at $24 \cdot 248 = 5\,952$, once the descent is replaced by freezing a core of twelve disjoint classes, pooling the monomial images disjoint from all of it and solving maximum clique on that pool. A subset of a class is again a class, so where a family is not exact the repeats are simply dropped.

What the layer costs in axis points is second order, and a rotation recovers part of it. The condition is $|\langle a, w\rangle| \le 1/\sqrt{2}$, and what fails it is an axis point lying along a frame direction: of the axis of [34], carried into the frame's coordinates, 16 of 24 survive in dimension 28 and 32 of 40 in dimension 29, where the eight omitted lines are at $|\langle a, w\rangle| = 1$ exactly. A maximum 60° code is a rotated copy of the root system and the rotation is free, so in dimensions 30 and 31 the axis is turned before it is carried across — by 45°, 45° and 90° in three coordinate planes, which keeps every coordinate an integer quadruple $(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6})/24$ — and 68 of 72 and 118 of

126 survive, against 48 and 110 unrotated. Hence

$$192\,592 + 3 \cdot 3\,968 + 16 + 384 = 204\,896,$$

$189\,616+19\,840+32+480 = 209\,968, \quad 184\,656+35\,712+68+576 = 221\,012,$

the middle count having 12 zero-sum triangles and 2 antipodal pairs, since $\tau(5) = 40$ is not a multiple of 3 and the 40 directions of a maximum $60^\circ$ code in $\mathbb{R}^5$ sum to zero, so 13 disjoint triangles cannot be carved out of one.

*Dimension 31: the same layer, and a density.* Here $96k = 672$, and the layer is available once the 42 type-$B$ classes hold at least 10 251 of the 10 416 lines that 42 full classes give — 98.4 per cent. The descent above reaches 10 183, and for a while that looked like the end of it. It is not an existence question: two uniform monomial images of the 248-line class are disjoint with probability $p = 0.4319$, so a family of 42 is a clique in a graph of that density on the $\sim 10^{12}$ images, where the random-graph clique number $2\ln(10^{12})/\ln(1/p)$ is about 66. It is a question about what a search that can only *sample* can reach: it places the $k$-th class while $np^{k-1} \geq 1$ and therefore stops at $k = 1 + 12\ln 10/\ln(1/p)$, which is 34 at $p = 0.4319$, so forty-two needs $p \geq 10^{-12/41} = 0.5097$. The density is a design choice. Writing $S_i$ for the set of 44 octads supporting class $i$, the probability that two images are disjoint falls with $|S_i \cap S_j|$, and

$$\sum_{i<j} |S_i \cap S_j| = \sum_O \binom{n_O}{2}, \qquad \sum_O n_O = 42 \cdot 44 = 1\,848$$

over the 759 octads, so a flat octad coverage gives a stronger guarantee than additional further sampling: the sum is 1 898 for the family the uniform scan produced and at least 1 419 for any 42 supports at all. Screening candidates on the free slots their octads still have, rather than on what they cost now, gives a family at 1 670 with $p = 0.5467$ and 10 328 of the 10 416 owner lines, and

$$175\,904 + 61\,968 + 118 + 672 = 238\,662.$$

How far this can go is known to different depths. In all four dimensions the layer is at the ceiling (15) — every vector of a frame on every direction of a cross-polytope, and 24 head lines is the most $\mathbb{R}^{24}$ holds. So is the axis, and exhaustively: a further axis point would be a *unit* vector of $\{a : \langle a, a_i\rangle \leq \frac{1}{2},\ \langle a, z\rangle \leq \sqrt{3}/2,\ |\langle a, w\rangle| \leq 1/\sqrt{2}\}$, whose right-hand sides are all positive, so that set is a polytope with the origin interior and the convex $|a|$ attains its maximum at a vertex; enumerating the vertices of all four gives 0.7654, 0.7929, 0.8165 and 0.8660, and none of the four axes admits another point at all. The direction weight $w = \sum_i (|Z_i| - 1)$ is at $\lfloor 2\tau(k)/3 \rfloor$, which is a theorem: two classes sharing a direction would meet at $\langle u, u'\rangle \leq 1$ throughout and so be one class, so the parts are disjoint; a part is pairwise at $\leq -\frac{1}{2}$ and $|\sum z|^2 \geq 0$ holds it to three; across parts the cosines are $\leq \frac{1}{2}$; hence $\bigcup Z_i$ is a $60^\circ$ code, $3t + 2q \leq \tau(k)$, and the maximum of $2t + q$ over that is $\lfloor 2\tau(k)/3 \rfloor$, attained here as 16, 26, 48 and 84. Only $\tau(4) = 24$ is

proved, so only dimension 28 reaches an unconditional ceiling: against the proved $\tau(5) \le 44$, $\tau(6) \le 77$ and $\tau(7) \le 134$ the unconditional weights are 29, 51 and 89, contributing 1 488, 1 488 and 2 480 points to anyone who improves a low-dimensional kissing number. What is otherwise left is the class problem of §6.2, at +32, +52, +96 and +168 for every extra line of the 248-line independent set, against the Delsarte ceiling of 425.45 lines; and in dimension 31 also the packing, where 88 repeated lines can contribute up to +352.

Three packages verify the four — `dim28-norm8-frame-layer`, `dim29-30-frame-layer` and `dim31-frame-layer`. Each rebuilds the 196 560 minimal vectors from the Golay code and checks the inner-product signature of the 98 280 lines before trusting them, carries every direction and axis point as an exact algebraic number — an integer quadruple $(a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6})/24$ for the frame layer — and decides every pair exactly: integer inner products in $\mathbb{R}^{24}$, and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ in $\mathbb{R}^k$. Each checks that the classes are disjoint sets of minimal vectors with pairwise inner products at most 1, that the heads are lattice vectors of the stated squared length, that no equator point is deleted, and that the four families have pairwise distinct $(|x|^2, |y|^2)$, so that the count adds no point twice.

F.6. **Dimension 38: large codimension.** For $k \le 7$ the number of parts of $Z$ binds in (4), and a part has maximum contribution when it is a triple: the number of triples is at most $\lfloor K(k)/3 \rfloor$, which is $0, 2, 4, 8, 13, 24, 42$, so at most $42 \times 248$ lines carry caps; one therefore takes the independent sets as large as possible. At $k = 5$ the bound 13 is not attained — the 40 directions of a maximum 60° code in $\mathbb{R}^5$ sum to zero, so thirteen disjoint zero-sum triples would leave a fortieth direction summing to zero on its own — but the weight $\sum_i (|Z_i| - 1) = 26$ is, by twelve triples and two antipodal pairs. At larger $k$ the parts of $Z$ outnumber the disjoint maximum independent sets available, and the binding question becomes how much of the shell admits a partition into independent sets, a colouring question.

**Theorem F.5.** $K(38) \ge 591\,612$.

*Proof sketch.* Take $L = \Lambda_{24}$, $k = 14$ and $t = \frac{2}{3}$. For $Z$ take the 1 932-point kissing configuration of $\mathbb{R}^{14}$, which admits a partition into 644 zero-sum triples. The 98 280 minimal lines of $\Lambda_{24}$ admit a partition into 644 pairwise disjoint independent sets, so every line carries caps and the equator is empty: (4) gives $196\,560 - 2 \cdot 98\,280 + 2 \cdot 98\,280 \cdot 3 = 6 \cdot 98\,280 = 589\,680$, the ceiling $3K(24)$ of the family.

The pole set is non-empty even though $Z$ already occupies every direction of the 1 932-point code. By (9) a pole needs 30° of separation from the directions; the covering radius of that code is 54.7°. There is a rotated copy of the code no point of which lies within 30° of $Z$, so all 1 932 of them are poles and the total is $589\,680 + 1\,932 = 591\,612$, which is exactly the family's ceiling $3K(24) + K(14)$.

Such a rotation is not found by sampling. A $30^\circ$ cap has relative measure $1.50 \times 10^{-5}$ on $S^{13}$, so the 1 932 caps cover at most 2.89% of it; a random rotation accordingly loses 56 points on average, with standard deviation 11, and an independent-points model gives $e^{-56}$ for a clean one. It is found instead by descending on $\mathrm{SO}(14)$. Scale the $w_i$ to $|w_i|^2 = 8$, so that the separation asks for $\langle Qw_i, w_j\rangle \le 4\sqrt{3}$; the objective $f(Q) = \sum_i \mathrm{softplus}\big(\mathrm{smax}_j \langle Qw_i, w_j\rangle - 4\sqrt{3}\big)$ is then differentiable, smax being a smooth maximum, and the geodesic step $Q \mapsto \exp(-\eta\,\mathrm{skew}(\nabla f\, Q^{\mathsf T}))Q$ keeps the iterate on the group. The real solution is then made rational by the Cayley transform, $(I-S)(I+S)^{-1}$ being rational and orthogonal for every rational skew-symmetric $S$, so the verification stays exact: $Q = N/D$ with $N$ integral, $NN^{\mathsf T} = D^2 I$ and $|\langle Nw_i, w_j\rangle| \le \mathrm{isqrt}(48D^2)$ for all $1\,932^2$ pairs. □

The partition into independent sets is the substantial step. Let $b(x,y) = \langle x, y\rangle \bmod 2$. Since $L$ is even, $b$ descends to an alternating bilinear form on the $\mathbb{F}_2$-space $L/2L$. For a totally isotropic subspace $R \subset L/2L$ of dimension 4 put

$$X_R = \{\, \pm u \;:\; u \in \Lambda,\ b(u,x) = 0 \text{ for every } x \in R \,\},$$

which for the subspaces used here contains 6 120 of the 98 280 lines. Every known maximum independent set of $\Gamma_\gamma(\Lambda_{24})$ lies inside such a set. Local search restricted to one $X_R$ returns independent sets of 230 to 244 lines, whereas the same search on the whole graph returns 152. The explicit partition is in [26].

F.7. **Dimension 39.** In dimensions 38 and 39 the chain that maximises the total has three levels, $(n, 8, 2)$, and reads

$$K(n) \;\ge\; A\big(n, \lceil n/4\rceil\big) + 128\, A(n,8,8) + 4\binom{n}{2}.$$

**Theorem F.6.** $K(39) \ge 756\,116$.

*Proof.* Echols [18] improves $A(39,8,8)$ from 3 323 to 3 324; Brouwer's table [4] carries the new value. With $A(39,10) \ge 327\,680$ [52],

$$327\,680 + 128 \cdot 3\,324 + 4 \cdot 741 \;=\; 327\,680 + 425\,472 + 2\,964 \;=\; 756\,116. \qquad \square$$

The gain is $128\,(3324-3323) = 128$. Echols [18] computes kissing numbers from the fixed level-one length $n_0 = 32$, which in dimension 39 falls below the record; that paper therefore reports no value there. The chain optimises at $n_0 = n$ in dimensions 38, 39, 40, 43 and 44; among these, 38 and 39 are the two whose totals exceed the published entry; §F.6 supersedes dimension 38. That entry is 755 988, which [6] attributes to the Edel–Rains–Sloane construction; of the three dimensions named in [46] it is dimension 43, not dimension 39, whose entry that table credits to the antipode construction. That paper also reconstructs the twelve cross-section packings of 1982 in dimensions 38–47 and computes their kissing numbers, reporting that they

exceed the best previously tabulated lower bounds in dimensions 42–47. Dimensions 38 and 39 are the only two this paper claims inside that span of 38–47, and neither lies in 42–47.

The verification in [26] checks the constant-weight code over all $\binom{3324}{2}$ pairs and the ten geometric conditions of the chain as exact rational inequalities, three of them tight. The value $A(39, 10) \geq 327\,680$ is taken from [52] and is not verified here; the existing record 755 988 rests on the same value.

F.8. **September attribution.** The frozen manuscript described its original 53-entry scope as follows.

*The fifty-three listed below.* We have found no earlier published claim in dimensions 18, 25–31, 38, 39, 49–63, 68–71 or 73–96. The one-point distribution of the minimal shell of $\Gamma_{72}$ follows from Venkov's theorem in a few lines; the bound in dimension 71, which is that distribution's $n[0]$, may therefore already be known to specialists, though we have found no record of it. Dimensions 62, 63 and 96 carry no new mathematics either. They evaluate the construction of [19] at three lengths where we find no record of its having been evaluated: its seven worked examples are dimensions 32, 36, 40, 44, 64, 80 and 128. All three improve whenever the code tables do. The range "32 to 128" of that paper's abstract is the span of those seven examples; it tabulates no value by dimension, and dimension 96 does not appear in it. Its own totals depend on the code tables in the same way, and it says in closing that it does not expect them to stand as those tables improve.

F.9. **Historical obstruction notes.** These notes concern the superseded dimension-25–27 constructions.

*The dimension-*25 *lens-head construction is bounded by* 197 623*.* Its equator is fixed, and every head removes exactly one equator point. Two heads can never share a removal: a head $x$ that removes the equator point $p$ has $\langle x, p\rangle \geq (18 + \sqrt{30})/7 = 3.35389\ldots$, which one spherical 7-design proves, whereas two compatible heads sharing a removal would both need $\langle x, p\rangle \leq \frac{10}{3}$; over the class heads of the shipped configuration the smallest such inner product is 3.36603. The count $196\,560 + H + 3$ of §F.2 is therefore exact, and only $H$ is at issue. The class-head core is a union of blocks, and over the entire norm-6 shell of 16 773 120 vectors it is capped at 971 heads. The supplement is an independent set in one explicit graph, on the complete pool of 1 188 candidate heads over 1 188 distinct owners, whose density is 0.1538; constraint programming bounds its independence number by 89. With the two poles and the one non-lattice equator point contributing 3, the construction cannot exceed $196\,560 + 971 + 89 + 3 = 197\,623$. The shipped configuration is within 44 of that, so a further gain in dimension 25 requires a head outside the block-plus-supplement family altogether. The instance does not close, so 89 is what a particular solve proves rather than the independence number itself; a fixed seed reproduces it, and other runs

of the same model have proved bounds as low as 88. Every such bound is valid, so the figure is a ceiling on the construction in any case.

*The second-layer choices in dimensions* 25 *to* 27 *fail at opposite ends of one structure.* Dimensions 26 and 27 carry a second cap layer above the first, whose heads have $|x|^2 = 3$, lie at height 1, carry one antipodal pair of directions and contribute $+1$ apiece compared with $+2$ for a first-layer head. A candidate is admitted only when *every* first-layer head blocking it is deleted, so what governs the choice is not how often blockers are shared on average but whether candidates with few blockers share those blockers with each other. Neither is it enough to evaluate one candidate at a time: the class heads, the free heads and the second layer compete for the same owners, and a change that loses in isolation can improve the total once the other two are allowed to move with it, which is how the configuration of §F.4 was found. Dimension 27 fails although the sharing is high. Its next admissible tier has 10 030 candidates carrying 252 582 incidences on 2 095 blocking heads, some 121 candidates apiece, yet each candidate carries about 25 blockers and needs all of them at once: deleting the 1 200 heads that block the most unlocks 253 candidates, and the choice becomes unfavorable only when the whole class layer is gone, with a loss of 4 190.

Dimension 25 has the opposite relative contribution and still gains nothing. Its lens heads are forced to height 1, since the poles require $y \le 1$ and a mirror pair $y \ge 1$ ; removing the two poles opens a second layer at $y = 2/\sqrt{3}$, $|x|^2 = 8/3$, $x = \frac{2}{3}v$ with $v \in \Lambda_{24}$ of norm 6. For such a head the condition $\langle x, z\rangle \le 2$ against the equator reads $\langle v, z\rangle \le 3$, and every norm-6 vector attains exactly 3, so the head removes nothing at all: it contributes $+2$ where a lens head contributes $+1$ because it removes an equator point. Deleting a lens head loses 1 and admitting a free head adds 2, the opposite of the ratio in dimensions 26 and 27. The joint improvement is nevertheless empty. Of 1.3 million norm-6 directions sampled against the shipped layer, the 654 blocked by at most ten lens heads are blocked by 630 distinct heads between them, barely one apiece. Here the blockers are few but private, and maximising $2|F| - |D| - 2$ over that instance is optimal at 0. Neither the relative contribution nor the average sharing settles the question; what matters is whether a set of candidates can be found whose blockers coincide.

Nor does dimension 25 admit the other kind of second layer, the one that carries dimensions 26 and 27: class heads at $|x|^2 = 8/3$, of the form $x = u + v/3$ with $u$ minimal and $|v|^2 = 6$, $\langle u, v\rangle = -3$. At $k = 1$ such a head carries both directions $\pm 1$, since two directions of one head need $\langle w, w'\rangle \le 1 - 2/h^2 = -\frac{1}{2}$ and $\mathbb{R}^1$ offers only $-1$; it therefore gives two cap points for the one equator point it removes, contributing $+1$, exactly as a lens head does, and it removes the two poles. Three of them would already improve on 197 579. None exists. Such a head must meet every lens head at $\langle x, x'\rangle \le 2 - 2/\sqrt{3}$, a cosine of 0.2989, and over 5.1 million of them, drawn from the blocks of as many norm-6 vectors, the smallest cosine attained against the shipped layer is 0.4034.

TABLE 6. The fifty-three improvements, grouped by construction; # is the number of dimensions in a row. The gain is $K_{\text{here}} - K_{\text{prev}}$ and the factor their ratio, computed dimension by dimension, with the factor attaining both its extremes at an end of the dimension range, to the two decimals shown; the largest gain need not, and in the 28–31 row it is dimension 30. "Previously" is [6] where that table has an entry and otherwise the value of §1.2. Dimensions 44 to 47 are absent, for the reasons given with Table 1, which gives every dimension.

| construction | dimensions | # | largest gain | factor | where |
|---|---|---|---|---|---|
| $\Gamma_{72}$ cross-sections | 68–71 | 4 | +2 271 920 766 | 1.09–7.85 | §B.8 |
| layers over $\Lambda_{24}$ | 25–27, 38 | 4 | +24 960 | 1.00–1.04 | §F.2–F.6 |
| second layers over $\Lambda_{24}$ | 28–31 | 4 | +572 | 1.00–1.00 | §F.5 |
| layers over $P_{48}$ | 49–61 | 13 | +7 481 540 | 1.00–1.14 | §C.4 |
| the ERS chain | 62, 63 | 2 | +86 001 280 | 1.36–2.64 | §D.1 |
| the ERS chain at 96 | 96 | 1 | +6 668 627 072 | 2.07–2.07 | §D.2 |
| layers over $\Gamma_{72}$ | 73–95 | 23 | +254 796 296 | 1.00–1.04 | §C.4 |
| constant-weight codes | 39 | 1 | +128 | 1.00–1.00 | §F.7 |
| the bent-coset hexagon | 18 | 1 | +704 | 1.09–1.09 | §E |
| total | | 53 | | | |

In Table 6 the factor is largest where the published table has no entry at all. Three rows exceed 2, and in none of them is the gain a new construction: the cross-sections count a classical configuration, and dimensions 62, 63 and 96 evaluate a construction of 1998 against the code tables of 2026. The rows with factors near 1 span the most dimensions, 46 of the 53, though the two largest absolute gains lie in rows whose factor exceeds 2.

## APPENDIX G. THE HISTORICAL RESULTS TABLE

Table 7: All fifty-three improvements. "Previously" is [6] where that table has an entry and otherwise the direct sum or inherited value of §1.2. Dimensions 44 to 47 are absent: [46] gives larger values in dimensions 44 and 45; dimensions 46 and 47 are already in the literature (§1.5).

| $n$ | previously | this work | construction |
|---|---|---|---|
| 18 | 7 654 | 8 358 | bent-coset hexagon over odd $\mathrm{BW}_{16}$ (§E) |
| 25 | 197 056 | 197 579 | layers over $\Lambda_{24}$, lens heads (§F.2) |
| 26 | 198 550 | 199 806 | layers over $\Lambda_{24}$, one hexagon of leans (§F.4) |
| 27 | 200 044 | 201 566 | layers over $\Lambda_{24}$, two coset triangles (§F.4) |
| 28 | 204 520 | 204 896 | second layer over $\Lambda_{24}$, norm-8 head (§F.5) |

| $n$ | previously | this work | construction |
|---|---|---|---|
| 29 | 209 496 | 209 968 | second layer over $\Lambda_{24}$, norm-8 frame (§F.5) |
| 30 | 220 440 | 221 012 | second layer over $\Lambda_{24}$, norm-8 frame (§F.5) |
| 31 | 238 350 | 238 662 | second layer over $\Lambda_{24}$, norm-8 frame (§F.5) |
| 38 | 566 652 | 591 612 | layers over $\Lambda_{24}$, $k = 14$ |
| 39 | 755 988 | 756 116 | constant-weight codes |
| 49 | 52 416 002 | 52 430 140 | layers over $P_{48}$ |
| 50 | 52 416 006 | 52 458 418 | layers over $P_{48}$ |
| 51 | 52 416 012 | 52 500 816 | layers over $P_{48}$ |
| 52 | 52 416 024 | 52 585 516 | layers over $P_{48}$ |
| 53 | 52 416 040 | 52 698 222 | layers over $P_{48}$ |
| 54 | 52 416 072 | 52 922 906 | layers over $P_{48}$ |
| 55 | 52 416 126 | 53 299 730 | layers over $P_{48}$ |
| 56 | 52 416 240 | 54 085 392 | layers over $P_{48}$ |
| 57 | 52 416 306 | 54 534 176 | layers over $P_{48}$ |
| 58 | 52 416 510 | 55 893 422 | layers over $P_{48}$ |
| 59 | 52 416 604 | 56 505 668 | layers over $P_{48}$ |
| 60 | 52 416 841 | 57 477 123 | layers over $P_{48}$ |
| 61 | 52 417 154 | 59 898 694 | layers over $P_{48}$ |
| 62 | 52 417 932 | 71 310 732 | ERS chain $(62, 15, 2)$ |
| 63 | 52 418 564 | 138 419 844 | ERS chain $(63, 15, 2)$ |
| 68 | 331 737 984 | 361 275 480 | cross-section of $\Gamma_{72}$, $k = 4$ |
| 69 | 331 737 984 | 627 822 180 | cross-section of $\Gamma_{72}$, $k = 3$ |
| 70 | 331 737 984 | 1 249 778 250 | cross-section of $\Gamma_{72}$, $k = 2$ |
| 71 | 331 737 984 | 2 603 658 750 | cross-section of $\Gamma_{72}$, $k = 1$ |
| 73 | 6 218 175 602 | 6 218 179 814 | layers over $\Gamma_{72}$ |
| 74 | 6 218 175 606 | 6 218 192 454 | layers over $\Gamma_{72}$ |
| 75 | 6 218 175 612 | 6 218 209 308 | layers over $\Gamma_{72}$ |
| 76 | 6 218 175 624 | 6 218 243 016 | layers over $\Gamma_{72}$ |
| 77 | 6 218 175 640 | 6 218 285 152 | layers over $\Gamma_{72}$ |
| 78 | 6 218 175 672 | 6 218 377 840 | layers over $\Gamma_{72}$ |
| 79 | 6 218 175 726 | 6 218 529 454 | layers over $\Gamma_{72}$ |
| 80 | 6 218 175 840 | 6 218 849 228 | layers over $\Gamma_{72}$ |
| 81 | 6 218 175 906 | 6 219 034 270 | layers over $\Gamma_{72}$ |
| 82 | 6 218 176 110 | 6 219 605 578 | layers over $\Gamma_{72}$ |
| 83 | 6 218 176 204 | 6 219 621 690 | layers over $\Gamma_{72}$ |
| 84 | 6 218 176 441 | 6 220 293 304 | layers over $\Gamma_{72}$ |
| 85 | 6 218 176 754 | 6 220 745 414 | layers over $\Gamma_{72}$ |
| 86 | 6 218 177 532 | 6 223 567 844 | layers over $\Gamma_{72}$ |
| 87 | 6 218 178 164 | 6 224 703 598 | layers over $\Gamma_{72}$ |
| 88 | 6 218 179 920 | 6 230 206 492 | layers over $\Gamma_{72}$ |
| 89 | 6 218 181 330 | 6 233 054 848 | layers over $\Gamma_{72}$ |
| 90 | 6 218 183 254 | 6 238 744 788 | layers over $\Gamma_{72}$ |
| 91 | 6 218 187 548 | 6 247 801 034 | layers over $\Gamma_{72}$ |

| $n$ | previously | this work | construction |
|---|---|---|---|
| 92 | 6 218 195 048 | 6 266 405 667 | layers over $\Gamma_{72}$ |
| 93 | 6 218 205 368 | 6 294 850 360 | layers over $\Gamma_{72}$ |
| 94 | 6 218 225 496 | 6 355 739 600 | layers over $\Gamma_{72}$ |
| 95 | 6 218 268 750 | 6 473 065 046 | layers over $\Gamma_{72}$ |
| 96 | 6 218 372 160 | 12 886 999 232 | ERS chain $(96, 24, 6, 1)$ |

## Use of AI assistance

The construction of §F.3 was found by Claude Opus 5 (Anthropic, maximum reasoning effort) in a chat session in which Alexey Kravatskiy provided the problem, Cohn's table [6] and Cohn's data set [7]. It appeared at the fortieth prompt; most of the thirty-nine earlier prompts were some form of "try harder".

The remaining constructions, the package [26] and this exposition were then developed by Kravatskiy with Claude Code (Anthropic), a coding agent built on the same model; those of the historical dimensions 25–27 with Claude Fable 5.1 in the same way, and those of the historical frame layer with Claude Code again, all three inside the joint work with H. Cohn and B. Lindow. GPT-6 Astra and GPT-6 Sol were used inside ChatGPT and in OpenAI Codex. Astra was used by H. Cohn to prepare the long internal summary of the dimensions 25–27 included in the correspondence; Codex was used in that collaboration for computational exploration and exact certification, including the four-triangle repartition and the later construction choices recorded in the correspondence. These tools also assisted with editorial review, code review, arithmetic cross-checks, implementation, verification runs and exposition; the authors set the mathematical direction, selected the lines to pursue, and reviewed every text, certificate and stated limitation.

In neither setting did the model predict success, and it said so in plain terms: "I have not produced a new SOTA … continuing would be me generating variations faster than either of us can check them."

## References


1. Christine Bachoc and Frank Vallentin, *New upper bounds for kissing numbers from semidefinite programming*, J. Amer. Math. Soc. **21** (2008), 909–924, `arXiv:math/0608426`.
2. Federico Bianchi, Yongchan Kwon, Aneesh Pappu, and James Zou, *Harnessing the collective intelligence of AI agents in the wild for new discoveries*, 2026, preprint, `arXiv:2606.10402`.
3. Peter Boyvalenkov and Danila Cherkashin, *The kissing number in* 48 *dimensions for codes with certain forbidden distances is* 52 416 000, Results Math. **80** (2025), no. 1, Paper No. 3, `arXiv:2312.05121`.
4. Andries E. Brouwer, *Bounds for binary constant weight codes*, `https://aeb.win.tue.nl/codes/Andw.html`, accessed 2026-08-26; $A(39, 8, 8) = 3324$ is credited there to Echols. The former address `www.win.tue.nl/~aeb/` now redirects here permanently.

5. Henry Cohn, *Spherical codes*, `https://cohn.mit.edu/spherical-codes/`, online catalogue of the best known spherical codes.
6. ———, *Table of kissing number bounds*, `https://cohn.mit.edu/kissing-numbers/`, current records checked 2026-10-24; historical comparisons also use the archived August and September 2026 snapshots.
7. ———, *Table of kissing number bounds*, data set, DSpace@MIT, `https://hdl.handle.net/1721.1/153312`, accessed 2026-08-20; the archived data set behind the web table.
8. Henry Cohn and Noam Elkies, *New upper bounds on sphere packings I*, Ann. of Math. (2) **157** (2003), no. 2, 689–714, `arXiv:math/0110009`.
9. Henry Cohn, Yang Jiao, Abhinav Kumar, and Salvatore Torquato, *Rigidity of spherical codes*, Geom. Topol. **15** (2011), no. 4, 2235–2273, `arXiv:1102.5060`.
10. Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska, *The sphere packing problem in dimension* 24, Ann. of Math. (2) **185** (2017), no. 3, 1017–1033, `arXiv:1603.06518`.
11. ———, *Universal optimality of the $E_8$ and Leech lattices and interpolation formulas*, Ann. of Math. (2) **196** (2022), no. 3, 983–1082, `arXiv:1902.05438`.
12. Henry Cohn and Anqi Li, *Improved kissing numbers in seventeen through twenty-one dimensions*, 2024, preprint, `arXiv:2411.04916`.
13. John H. Conway and Neil J. A. Sloane, *Sphere packings, lattices and groups*, 3 ed., Grundlehren der mathematischen Wissenschaften, vol. 290, Springer, New York, 1999.
14. David de Laat, Nando M. Leijenhorst, and Willem H. H. de Muinck Keizer, *Optimality and uniqueness of the $D_4$ root system*, 2024, preprint, `arXiv:2404.18794`.
15. Philippe Delsarte, *An algebraic approach to the association schemes of coding theory*, Philips Res. Rep. Suppl., no. 10, Philips Research Laboratories, 1973.
16. Philippe Delsarte, Jean-Marie Goethals, and Johan J. Seidel, *Spherical codes and designs*, Geom. Dedicata **6** (1977), no. 3, 363–388.
17. Ivan Dorofeev, Xiaoming Sun, and Chengu Wang, *Optimal extensions of cross-sections: sphere packings in dimensions* 38 *to* 43, 2026, preprint, version 4, `arXiv:2607.20359`.
18. William Echols, *New lower bounds for constant-weight codes via seeded bit-swap tabu search*, 2026, preprint, `arXiv:2608.13906`.
19. Yves Edel, Eric M. Rains, and Neil J. A. Sloane, *On kissing numbers in dimensions* 32 *to* 128, Electron. J. Combin. **5** (1998), Research Paper 22, 5 pp., `arXiv:math/0207291`.
20. Mikhail Ganzhinov, *Highly symmetric lines*, Linear Algebra Appl. **722** (2025), 12–37, `arXiv:2207.08266`.
21. Bogdan Georgiev, Javier Gómez-Serrano, Terence Tao, and Adam Zsolt Wagner, *Mathematical exploration and discovery at scale*, 2025, preprint, `arXiv:2511.02864`.
22. Markus Grassl, *Bounds on the minimum distance of linear codes and quantum codes*, `https://codetables.de`, accessed 2026-08-23; the codes $[62, 26, 16]$, $[63, 27, 16]$, $[64, 28, 16]$, $[96, 33, 24]$ and $[24, 9, 12]_4$.
23. Boon Suan Ho, *A new lower bound for the kissing number in* 19 *dimensions*, 2026, preprint, `arXiv:2603.10425`.
24. Kenz Kallal, Tomoka Kan, and Eric Wang, *Improved lower bounds for kissing numbers in dimensions* 25 *through* 31, SIAM J. Discrete Math. **31** (2017), no. 3, 1895–1908, `arXiv:1608.07270`.
25. Valentin Khrulkov, Andrey Galichin, Denis Bashkirov, Dmitry Vinichenko, Oleg Travkin, Roman Alferov, Andrey Kuznetsov, and Ivan Oseledets, *GigaEvo: an open source optimization framework powered by LLMs and evolution algorithms*, 2025, preprint, `arXiv:2511.17592`.
26. Alexey Kravatskiy, *Kissing numbers: verification packages for new lower bounds in dimensions* 18 *through* 96, software and data, `https://github.com/alexlegeartis/KissingNumbers`, 2026.

27. Alexey Kravatskiy, Valentin Khrulkov, and Ivan Oseledets, *ImprovEvolve: basin-hopping meets LLM-guided evolutionary search*, AI for Math Workshop at the 43rd International Conference on Machine Learning (ICML 2026), 2026, https://openreview.net/forum?id=nE1q0gRDNr, arXiv:2602.10233.
28. Nando Leijenhorst and David de Laat, *Solving clustered low-rank semidefinite programs arising from polynomial optimization*, Math. Program. Comput. **16** (2024), no. 3, 503–534, arXiv:2202.12077.
29. Vladimir I. Levenshtein, *On bounds for packings in n-dimensional Euclidean space*, Soviet Math. Dokl. **20** (1979), 417–421.
30. Kevin Li, *Improved kissing numbers in nineteen and twenty-one dimensions*, https://hexagonmath.org/2610.00248v1, 2026, preprint, hexagon:2610.00248v1.
31. ———, *Improved kissing numbers in twenty-eight through thirty-one dimensions*, https://github.com/Felpix-Studios/kissing-number/tree/b823640f4e98134b9d55b27288132ee954a8affa/preprints/leech-lift-d28-d31, 2026, preprint and exact certificates; accessed 2026-10-24.
32. Qihong Lin, *A kissing configuration of* 352320 *points in* $\mathbb{R}^{32}$ *and* $a(32, 8, 8) \geq 1713$*: explicit certificates*, 2026, data set, version 1.1.0.
33. ———, *New lower bounds for kissing numbers in dimensions* 33 *to* 39*: explicit certificates*, 2026, data set, version 1.0.0.
34. Chengdong Ma, Théo Tao Zhaowei, Pengyu Li, Minghao Liu, Haojun Chen, Zihao Mao, Bo Li, Yuan Cheng, Yuan Qi, and Yaodong Yang, *Finding kissing numbers with game-theoretic reinforcement learning*, 2025, preprint, arXiv:2511.13391.
35. Fabrício C. Machado and Fernando M. de Oliveira Filho, *Improving the semidefinite programming bound for the kissing number by exploiting polynomial symmetry*, Exp. Math. **27** (2018), 362–369, arXiv:1609.05167.
36. Hans D. Mittelmann and Frank Vallentin, *High-accuracy semidefinite programming bounds for kissing numbers*, Exp. Math. **19** (2010), 175–179, arXiv:0902.1105.
37. Oleg R. Musin, *The kissing number in four dimensions*, Ann. of Math. (2) **168** (2008), 1–32, arXiv:math/0309430.
38. Gabriele Nebe, *An even unimodular* 72*-dimensional lattice of minimum* 8, J. Reine Angew. Math. **673** (2012), 237–247, arXiv:1008.2862.
39. ———, *A fourth extremal even unimodular lattice of dimension* 48, Discrete Math. **331** (2014), 133–136, arXiv:1312.3780.
40. Gabriele Nebe and Neil J. A. Sloane, *Table of the highest kissing numbers presently known*, http://www.math.rwth-aachen.de/~Gabriele.Nebe/LATTICES/kiss.html, last revised February 2011; the page now points to [6].
41. Alexander Novikov, Ngân Vũ, Marvin Eisenberger, Emilien Dupont, Po-Sen Huang, et al., *AlphaEvolve: a coding agent for scientific and algorithmic discovery*, 2025, preprint, arXiv:2506.13131.
42. Andrew M. Odlyzko and Neil J. A. Sloane, *New bounds on the number of unit spheres that can touch a unit sphere in n dimensions*, J. Combin. Theory Ser. A **26** (1979), 210–214.
43. Antoine Olivier, *Kissing number in dimension* 37*:* $\tau_{37} \geq 500008$*, built on the code of Qihong Lin*, 2026, data set.
44. Michio Ozeki, *A numerical study of Siegel theta series of various degrees for the* 48*-dimensional even unimodular extremal lattices*, Tsukuba J. Math. **40** (2016), 139–186.
45. Kurt Schütte and Bartel L. van der Waerden, *Das Problem der dreizehn Kugeln*, Math. Ann. **125** (1953), 325–334.
46. Xiaoming Sun and Chengu Wang, *Sphere packings and kissing numbers in dimensions* 39*,* 43*, and* 45 *from the antipode construction*, 2026, preprint, version 3, arXiv:2607.20359.
47. Rustem Takhanov, Zhenisbek Assylbekov, and Stanislav Yun, *Structure of kissing arrangements in* $\mathbb{R}^{12}$ *and a place for the* 841*st sphere*, 2026, preprint, arXiv:2606.18984.

48. Rustem Takhanov and Stanislav Yun, *New lower bounds for kissing numbers in dimensions* 25*,* 26*,* 27*,* 29 *and* 31, `https://github.com/k-nic/kissing_d25-31`, 2026, preprint, `arXiv:2609.21591`; exact certificates accessed 2026-10-24.
49. Boris B. Venkov, *Even unimodular extremal lattices*, Trudy Mat. Inst. Steklov. **165** (1984), 43–48.
50. Maryna S. Viazovska, *The sphere packing problem in dimension* 8, Ann. of Math. (2) **185** (2017), no. 3, 991–1015, `arXiv:1603.04246`.
51. Shuxing Yang, Rui Zhao, Junyao Wu, Yize Wang, Fujia Chen, Kaihao Zhu, Wenhao Li, Zichen Li, Yaqi Li, Shenzhan Hong, Yuang Pan, Junjie Yang, Taowen Deng, Jincheng Mi, Hongsheng Chen, and Yihao Yang, *Large-scale autonomous discovery of kissing number constructions*, 2026, preprint, version 1, `arXiv:2609.35051`.
52. Victor A. Zinov'ev and Simon N. Litsyn, *On shortening of codes*, Problems Inform. Transmission **20** (1984), no. 1, 1–7.
53. Victor A. Zinoviev and Thomas Ericson, *New lower bounds for contact numbers in small dimensions*, Problems Inform. Transmission **35** (1999), 287–294.

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